Pell residues instead of power values

Explicit Diophantine consistency equations
in 395 nonnegative integers and 663 unrestricted integers

Prepared for Vladimir Reshetnikov's ProveIt project

AI-assisted mathematical research with reproducible exact algebra

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Abstract

We construct literal integer-coefficient quartics P_{395} and Q_{663} satisfying

$$\mathrm{ZFC} \vdash \big(\nexists x \in \mathbb{N}^{395}\ P_{395}(x) = 0\big) \iff \mathrm{Con(ZFC)} \iff \big(\nexists z \in \mathbb{Z}^{663}\ Q_{663}(z) = 0\big).$$

The natural equation has 321 quadratic residuals, 7,323 expanded monomials, and coefficient height 83,842,758. A 687-variable integer companion retains smaller coefficients. Complete residual lists and fully collected polynomial files are supplied.

The central new transformation replaces eleven full power values $(q+1)^n$ by residues of indexed Pell solutions. A single congruence makes $q+1$ a root of the common Pell characteristic polynomial modulo q^{H+1}. Separating the growth lower bound from the actual Pell base makes that congruence compatible with completeness. Exact lower digits and binomial divisibility are then recovered without representing the full powers. This reduces the underlying natural arithmetic graph from 690 to 655 coordinates. Combining it with previously established signed Pell-index arguments and exact sign inference gives the integer reduction.

The comparison is against the current incoming 407/888/917-variable direct quartics, not against obsolete milestones. These constructions encode the calibrated numerical proof calculus C_{222} directly; they do not simulate K215 or instantiate Jones's universal polynomial. Separate twelve-integer refinements improve the few-variable tradeoff. We claim explicit improvements in stated measures, not global minimality or worldwide priority. The proof is ordinary mathematics formalizable in ZFC, composed with the identified inherited calculus calibration. No new end-to-end Lean/Rocq proof term or emitted object-language ZFC derivation is claimed.

Contents

1 Results, comparison, and attribution **3**
1.1 What is new here . . . 3
1.2 What “simpler” means in this report . . . 3

2 The calibrated source and why it expresses refutability **4**
2.1 A bounded certificate with eight columns . . . 4

3 The arithmetic interface preserved by the new construction **5**
3.1 Factorials and the finite product graph . . . 5
3.2 Why the radix bound is not circular . . . 5

4 A Pell base with a separately certified growth bound **6**

5 Replacing eleven full powers by one common residue condition **6**
5.1 Exact digits . . . 7
5.2 Divisibility without the digit value . . . 7
5.3 Why this saves witnesses . . . 7

6 Signed auxiliary Pell witnesses: the inherited theorem **8**

7 Integer domains, sign certificates, and exact projection **8**
7.1 Which quotient signs are unnecessary . . . 8
7.2 Encoding only the necessary signs . . . 9
7.3 Projection and the single quartic . . . 9

8 The internal-ZFC statement and its proof boundary **9**

9 Two additional twelve-integer improvements **10**

10 Reproduction, inspection, and remaining research **10**

A The complete 395-variable natural equation **12**

B The complete 663-variable integer equation **20**

C The complete 687-variable small-coefficient equation **29**

1 Results, comparison, and attribution

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$; an integer variable has no sign restriction. A polynomial's height is the maximum absolute coefficient after full expansion and collection of equal monomials.

Theorem 1.1 (Fixed ordinary equations)**.** *The complete residual lists in Appendices A, B, and C define*

$$P_{395}(x) = \sum_{j=0}^{320} R_j(x)^2, \qquad Q_{663}(z) = \sum_{j=0}^{396} S_j(z)^2, \qquad Q_{687}(w) = \sum_{j=0}^{420} T_j(w)^2.$$

Each polynomial has exact degree four. In ZFC, existence of a zero of each polynomial over its stated domain is equivalent to existence of a finite ZFC refutation. Consequently its no-zero sentence is equivalent, provably in ZFC, to Con(ZFC)*.*

The term "fixed" excludes an unspecified machine index, a free source parameter, or a coefficient chosen using an undecidable predicate. Every coefficient and every variable is present in the delivered JSON. The appendices specify the three principal quartics without any executable macro. Powers, factorials, and bounded quantifiers below describe their construction; they are absent from the final polynomial language.

Status	Domain	Variables	Rows	Degree	Monomials	Height
Incoming	$\mathbb{N}$	407	323	4	8,420	83,843,774
This work	$\mathbb{N}$	395	321	4	7,323	83,842,758
Incoming	$\mathbb{Z}$	888	470	4	10,156	2,445,790,974,704
This work	$\mathbb{Z}$	663	397	4	7,268	2,445,790,974,704
Incoming	$\mathbb{Z}$	917	499	4	6,986	63,906,036
This work	$\mathbb{Z}$	687	421	4	5,602	63,906,013

Table 1: Like-for-like direct quartic comparisons against the latest inspected incoming archive. A row is a residual before squaring.

The comparison snapshot is ProveIt commit `ec80841c07e91be8956ce6ed4af9a9c0b782d3d9`. In particular, the incoming `ConZFC_407_888_Quartics_2026-10-24.zip` was inspected directly [2, 1]. The natural variable count decreases by 2.95%; the principal integer count decreases by 25.34%. The raw arithmetic graph has 655 coordinates and 581 quadratic rows, including 12 exact power requests, three exact binomials, eight binomial divisibility requests, and 15 independent Pell-index certificates.

1.1 What is new here

The new arithmetic step is the common modular-root construction in Sections 4–5: freely increase the Pell base above its growth lower bound, force a specified root modulo a common power of the radix, and use Pell residues directly in digit tests. The explicit mixed-domain integration, emitted coefficient lists, and new twelve-integer centered circuits are also contributions of this iteration.

The numerical calculus, its ZFC calibration, accumulator argument, CRT compiler, factorial representation, recycled radix, additive Pell-index reuse, and natural contraction method are inherited from the 407/888 package and its dependencies. Crucially, the signed auxiliary-index theorem was already proved in `ConZFC_Shared_Pell_1873.zip`; it is reused with credit, and independently checked here [3]. Earlier sign-orbit and positivity arguments are also credited [4]. Rediscovering those facts during this investigation does not establish novelty. The archive preserves the applicable source notices, original input ZIP, exact source hashes, and mathematical reviews.

1.2 What "simpler" means in this report

The repository already contains equations with far fewer variables, including a 37-integer quartic using a different coefficient architecture, and nine-positive-variable equations of high degree [5, 1]. Our main results improve the *direct quartics with moderate literal coefficients and completely collected expansions*. They do not dominate every construction in the repository. Section 9 separately improves the twelve-integer degree/operation frontier. There is no theorem here that the global minimum number of variables, degree, or description length has been found.

2 The calibrated source and why it expresses refutability

The source is the numerical calculus C_{222}. Its free truth code is 0 and its false target is 1. Define

$$I(a,b) = 2(a+b)(a+b+1) + 4b, \quad M(a,b) = I(a,b) + 1, \quad A(a,b) = I(a,b) + 2, \quad E(a,b) = I(a,b) + 3,$$

and $N(a) = I(a,1)$, $X(a,b) = N(A(a,N(b)))$. The constructors encode implication, membership, universal quantification, equality, negation, and existential quantification in the inherited arithmetic convention. Eight unconditional axiom families and one guarded bundle are fixed polynomial compositions of these constructors. Their complete definitions, the exact twelve branch rules, and the source identity checks are contained in the preserved parent archive, especially `upstream/provenance/upstream/calculus.tex`.

Each rule has two padded premises p, q and one conclusion v. The branches are truth padding $(0,0,0)$, the eight axioms $(0,0,c_j)$, the guarded bundle $(0,0,c_*)$, modus ponens $(a, I(a,b), b)$, and generalization $(b, 0, A(a,b))$. The bundle guard $c \neq a, b$ is represented by

$$(c-a)^2(c-b)^2 = (d+1)^2.$$

The four schema parameters a, b, c, d are shared across the finite disjunction; only the selected branch must hold.

The exact inherited semantic endpoint is

$$C_{222}(1) \quad \longleftrightarrow \quad \exists p\ \mathrm{Prf}_{\mathrm{ZF}}(p, \bot) \quad \longleftrightarrow \quad \exists p\ \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot). \tag{1}$$

The named sources are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`, pinned to commit `08200eb10fde2db408c69dffd840d6920a569b1f` in the parent provenance. The second equivalence uses the constructible-universe equiconsistency of ZF and ZFC. The first is a calculus theorem before the machine-specific endpoint: this construction uses no transition table. Our ordinary proof composes with these identified calibration results; the complete transitive Lean development has not been recompiled here.

2.1 A bounded certificate with eight columns

For a sufficiently padded horizon $t = t_0 + 1$, let

$$M = M_0 + 1 = \left((t+1)^t\right)^{2^{22}}.$$

The parent label-majorant certificate shows that this carrier is large enough for every label generated by the bounded schema parameters. At row $i < t$, an accumulator update has the form

$$S'(1 + vM) = S(1 + pM)(1 + qM). \tag{2}$$

Project the rule's labels p, q, v to their branch-specific polynomial expressions. The only locally bounded columns are then

$$a, b, c, d, \ \ S, S', u, u'.$$

The first four have bound t, the next two bound y, and the last two bound x. The retained beta equations are

$$x = S + (1 + (i+1)y)u, \qquad x = S' + (1 + (i+2)y)u'.$$

Because $S, S' \leq y$, these are canonical remainder equations; consecutive rows decode the same shared state. The endpoint rows give $S_0 = M + 1$ and $S_t = 1$. Completeness chooses y sufficiently large and divisible by $t!$, then chooses x by CRT.

Multiplying (2) gives equality at M of products of linear factors. The carrier bound prevents carries and recovers their equality as polynomials. Unique factorization recovers label multiplicities; the inherited single-conclusion flow argument converts this certificate into a finite Horn derivation of 1. Conversely a finite derivation can be padded, scheduled, bounded, and encoded in this form. This is the unchanged source theorem, not an inference from polynomial test samples.

3 The arithmetic interface preserved by the new construction

Put $R = 2 + t + x + y + M_0$ and $H = R^{2^{23}}$. The exponent is a fixed integer: its implementation is 23 squarings. A checked monotone majorant for the local polynomial has exponent 6,893,253, below 2^{23}, and bounds the local residual sum and every decoded coordinate. Let $T = H!$ and

$$B = \prod_{i=0}^{t-1}(1 + (i+1)T).$$

The explanatory product is eliminated by a finite arithmetic graph. The CRT certificate has positive encoded column representatives $a_1, \dots, a_8$ and an index representative C, with

$$B \mid \binom{a_j}{u_j + 1}, \qquad B = 1 + (C+1)T, \qquad F(C, a_1, \dots, a_8) \equiv 0 \pmod{B}. \tag{3}$$

Here F is the nonnegative local rule-and-beta polynomial and u_j are the heterogeneous bounds just stated.

For soundness, choose a prime divisor p of $1 + (i+1)T$. Since $p \nmid H!$, we have $p > H$. The falling-factorial identity shows that $B \mid \binom{a_j}{u_j+1}$ supplies a residue $w_j \in [0, u_j]$ modulo p. The canonical index equation gives $C \equiv i \pmod{p}$. Thus $F(i, w) \equiv 0 \pmod{p}$, while the majorant gives $0 \le F(i, w) < p$; therefore $F(i, w) = 0$. Completeness uses the pairwise coprime moduli $1 + (i+1)T$ and finite CRT. The factorial $(u_j + 1)!$ is coprime to B and can be cancelled in the converse falling-factorial argument.

3.1 Factorials and the finite product graph

For $n \ge 1$, write

$$\rho_n = (n+2)^{n+2}, \qquad V_n = (\rho_n + 1)^n, \qquad c_n = \binom{\rho_n + n}{n}.$$

The inherited elementary identity is

$$n! = \lceil V_n / c_n \rceil . \tag{4}$$

Indeed $V_n/c_n = n! \prod_{i=1}^{n}(\rho_n + 1)/(\rho_n + i)$ lies in $(n! - 1, n!]$. The estimate follows by bounding $1 - \prod(1 - \alpha_i)$ by $\sum \alpha_i$, and then using $n! n(n-1)/(2(\rho_n + 1)) < 1$. The exact quadratic graph uses $V_n + c_n = c_n f + r + 1$ and $c_n = r + s + 1$.

Set $Q = (1 + (t+1)T)^t + 2$ and $D = TQ - 1$. The product graph uses

$$B = \left(T^t t! \binom{Q + t}{t} \right) \bmod D. \tag{5}$$

The numerator equals $\prod_{i=1}^{t}(TQ + iT)$ and hence reduces to the desired product modulo D; the product is strictly smaller than D. These identities explain the three exact binomial requests and two factorial requests in the actual graph.

3.2 Why the radix bound is not circular

Choose the inherited recycled radix

$$q = B(V_H + 1), \qquad \sum_{j=1}^{8} a_j \le 8B. \tag{6}$$

Soundness first recovers all twelve retained exact powers from the growth frame and indexed-Pell criteria of Section 4. The modulus row for the exact power q^H forces $q > 0$. Thus (6), with the recovered $V_H \ge 1$, gives $B \ge 1$ and $q \ge 2$. The common Pell residue identity in Lemma 5.2 now follows without using any binomial or factorial interpretation. Only after these steps do the following digit estimates recover the factorials and the CRT product.

Only the coefficients through the requested digit must be less than q. For the factorial binomials, unimodality and the product formula give

$$\binom{\rho_n + n}{j} \le \binom{\rho_n + n}{n} \le V_n \qquad (0 \le j \le n).$$

First this proves the H-factorial digit and $T = H!$, then the t-factorial. For the remaining binomials, use only the raw remainder inequality $B < TQ$, without presuming (5) is already valid. With $e = t(H+1)$ one obtains

$$Q + t \le H^{e+1}, \qquad a_j < 8TQ \le H^{(t+2)(H+1)}.$$

Their lower indices k satisfy $k \le R$ and $k(t+2) \le H$. Hence

$$\binom{n}{j} \le n^k \le H^{H(H+1)} < V_H < q \qquad (0 \le j \le k)$$

for all remaining requests. Now the product binomial and all eight range tests are justified. The order matters: using the intended product value of B earlier would be circular. The complete inherited estimates are preserved in `proofs/inherited_radix_and_pell.txt`.

4 A Pell base with a separately certified growth bound

For $A \ge 2$, define

$$X_0 = 1,\ Y_0 = 0, \qquad X_{j+1} = AX_j + (A^2-1)Y_j, \quad Y_{j+1} = X_j + AY_j.$$

Then (X_j, Y_j) are the nonnegative solutions of $X^2 - (A^2-1)Y^2 = 1$. Their recurrence, classification, and index criterion are the standard Pell–Matiyasevich tools used by the parent construction [7, 8].

Let $\mathcal{A}$ be the finite set of distinct atoms occurring as a base, exponent, or output of the *12 retained exact-power requests*. Put

$$W = 2 + \sum_{a \in \mathcal{A}} a, \quad Z = z + 1, \qquad L^2 - \big((W+1)^2 - 1\big)(WZ)^2 = 1, \qquad A = L + \sigma, \tag{7}$$

with all displayed coordinates nonnegative. The parent put A directly in this growth norm. Introducing L separates a lower bound from the actual base.

Lemma 4.1 (Growth and cofinal choice)**.** *Every solution of (7) satisfies $A > W^W$. Conversely, after fixing the semantic atoms, a suitable L, Z exist and every integer $A \ge L$ is admissible.*

Proof. Pell classification gives $L = X_e(W+1)$ and $WZ = Y_e(W+1)$. The recurrence modulo W gives $Y_e(W+1) \equiv e \pmod W$. Thus $W \mid e$ and $e > 0$, so $e \ge W$. Consequently $L \ge (W+1)^e \ge (W+1)^W > W^W$. For completeness take $e = W$ and $Z = Y_W(W+1)/W$, then set $\sigma = A - L$. □

For a retained exact-power request (n, k, m), the main indexed pair at A and modulus $T_n = 2An - n^2 - 1$ still give

$$X_k + (n - A)Y_k \equiv m \pmod{T_n}. \tag{8}$$

The frame implies $0 \le m < T_n$ and $0 < n^k < T_n$; the usual Pell recurrence gives the same congruence with n^k in place of m. Therefore $m = n^k$. The modulus row with $T_n \ge 0$ excludes a zero base before any factorial or binomial interpretation is used. All exact exponents are positive by their source expressions.

The new modular-only upper indices are *not* required to be at most W. They retain their explicit indexed-Pell constraints $Y \ge k \ge 1$. This distinction saves additional frame coordinates without weakening any growth estimate used for an exact power.

5 Replacing eleven full powers by one common residue condition

Let $q \ge 2$, $b = q + 1$, and $M_q = q^{H+1} = q\,q^H$. The exact denominator q^H is already needed for the H-factorial digit, so forming M_q needs one multiplication. Add

$$\boxed{2A(q+1) = (q+1)^2 + 1 + M_q h_0.} \tag{9}$$

Lemma 5.1 (Compatibility with completeness)**.** *For fixed q, H there are arbitrarily large integers A satisfying (9). They can be chosen above L and $q+1$, with $h_0 \ge 0$.*

Proof. Since $\gcd(q+1, M_q) = 1$, the gcd of $2(q+1)$ and M_q is either 1 or 2. In the even case $q+1$ is odd, so $(q+1)^2+1$ is even. The linear congruence is therefore soluble. Its solution class contains arbitrarily large positive representatives. Choose one large enough; the resulting quotient is nonnegative. □

Lemma 5.2 (Pell residue identity)**.** *Under* (9)*, for every* $n \in \mathbb{N}$*,*

$$\boxed{\mathcal{R}_n = X_n(A) + (q+1-A)Y_n(A) \equiv (q+1)^n \pmod{q^{H+1}}.} \tag{10}$$

Moreover $\mathcal{R}_n > 0$*.*

Proof. In $\mathbb{Z}[s]/(s^2-(A^2-1))$ we have $(A+s)^n = X_n + Y_n s$. Equation (9) makes $s \mapsto q+1-A$ a homomorphism to $\mathbb{Z}/M_q\mathbb{Z}$, since $(q+1-A)^2 - (A^2-1) = (q+1)^2 - 2A(q+1) + 1$ vanishes there. This proves the congruence. For $n > 0$, the norm identity yields

$$X_n^2 - (A-1)^2 Y_n^2 = 1 + 2(A-1)Y_n^2 > 0,$$

so $X_n > (A-1)Y_n$ and $\mathcal{R}_n > qY_n > 0$. For $n = 0$, $\mathcal{R}_0 = 1$. □

5.1 Exact digits

For a requested lower index $1 \le k \le H$, set $d = q^k$. The new exact-digit rows are

$$X_n + (q+1)Y_n = AY_n + r + dc + qdh, \qquad d = r + s + 1, \qquad q = c + o + 1. \tag{11}$$

Thus $0 \le r < d$ and $0 \le c < q$. Because $qd \mid M_q$, Lemma 5.2 identifies $r + dc$ with the least residue of $(q+1)^n$ modulo qd. The binomial theorem and the lower-digit bounds of Section 3 force $c = \binom{n}{k}$. Conversely those correct low digits provide the row: the quotient $(\mathcal{R}_n - r - dc)/(qd)$ is integral, and is nonnegative because $\mathcal{R}_n > 0$ and $0 \le r + dc < qd$.

5.2 Divisibility without the digit value

For a range test only $B \mid \binom{n}{k}$ is required. Since $B \mid q$, we have $Bd \mid M_q$. Replace the exact-digit rows by

$$X_n + (q+1)Y_n = AY_n + r + Bdh, \qquad d = r + s + 1. \tag{12}$$

The least residue of $(q+1)^n$ modulo Bd is

$$r_0 + d\Big(\binom{n}{k} \bmod B\Big), \qquad 0 \le r_0 < d.$$

Higher digits vanish modulo B because $B \mid q$. Consequently (12) holds exactly when $B \mid \binom{n}{k}$. Again the quotient is nonnegative when constructed from the true residue, by positivity of $\mathcal{R}_n$.

5.3 Why this saves witnesses

The parent represented the entire numerator $(q+1)^n$ for each of the three exact and eight divisibility requests, then extracted its low digits. The new construction retains the already required indexed Pell pairs and their exact denominators, but removes those eleven numerator power graphs. The full-power count falls from 23 to 12. The new lower frame and root congruence cost a small common amount, yielding the net raw reduction $690 \to 655$. The final natural contraction is $655 \to 395$.

The order of witness selection is explicit: choose a finite source certificate, its true arithmetic values and radix; choose W, L; choose A in the soluble root class above L; then choose common auxiliary Pell/CRT witnesses and digit quotients. There is no equation requiring a semantic output to be fixed before a quantity on which it depends has been chosen.

6 Signed auxiliary Pell witnesses: the inherited theorem

We restate the sign argument used in the integer compilation, with credit to the earlier incoming proof [3]. Its role here is to permit a smaller integer-domain implementation of the new residue construction.

Theorem 6.1 (Signed auxiliary index recovery)**.** *Let $A, b \geq 2$, $X, Y, V \in \mathbb{N}$, $1 \leq k \leq Y$, and $U, S, R \in \mathbb{Z}$. Suppose*

$$\begin{aligned} X^2 - (A^2-1)Y^2 &= 1, & U^2 - (A^2-1)V^2 &= 1, & V &> 0, \\ S^2 - (b^2-1)R^2 &= 1, & Y^2 &\mid V, & & \\ b &\equiv A \pmod{U}, & b &\equiv 1 \pmod{2Y}, & & \\ S &\equiv X \pmod{U}, & R &\equiv k \pmod{2Y}. & & \end{aligned}$$

Then $(X, Y) = (X_k(A), Y_k(A))$. In particular the conclusion applies to the delivered system, which retains the stronger moduli $4Y$.

Proof. Classify the nonnegative main pair as $(X_i(A), Y_i(A))$, where $i \geq 1$, and $(|U|, V)$ as $(X_m(A), Y_m(A))$, where $m \geq 1$. The standard divisibility argument gives $Y_i \mid m$ and $i \leq m$. For completeness, first $Y_i \mid Y_m$ implies $i \mid m$: divide the index by i, use the unit X_i modulo Y_i, and the strict increase of the smaller remainder ordinate. Writing $m = di$, expansion modulo Y_i^2 gives

$$Y_m \equiv dX_i^{d-1} Y_i \pmod{Y_i^2}.$$

Since $\gcd(X_i, Y_i) = 1$, the stronger condition $Y_i^2 \mid Y_m$ implies $Y_i \mid d$, hence $Y_i \mid m$.

Classify $(|S|, |R|)$ at base b with index $j \geq 0$. Congruence of the base and the possibly negative sign of S give

$$X_j(A) \equiv \pm X_i(A) \pmod{X_m(A)}.$$

The Pell abscissa congruence lemma implies $j \equiv \pm i \pmod{2m}$. One can derive its signed version from the usual positive congruence theorem by $X_{j+2m} \equiv -X_j \pmod{X_m}$. Equivalently fold j to $[0, m]$: for $m \geq 2$, $2X_{m-1} < X_m$ makes the smaller positive residues distinct even up to sign. The endpoint $i = m$ has residue zero and is separate; when $m = 1$, the already known $Y \mid m$ gives $i = k = Y = 1$.

Since $Y \mid m$, we obtain $j \equiv \pm i \pmod{2Y}$. The recurrence at $b \equiv 1 \pmod{2Y}$ gives $Y_j(b) \equiv j \pmod{2Y}$, so the possibly negative sign of R also gives $j \equiv \pm k \pmod{2Y}$. Thus $i \equiv \pm k \pmod{2Y}$. Both i and k belong to $[1, Y]$. Equality modulo $2Y$ forces $i = k$; in the opposite-sign case $i + k = 2Y$ forces $i = k = Y$. This proves the assertion. □

The argument preserves the sign of the *main* pair. It permits arbitrary auxiliary signs. Replacing S, R by their absolute values on the same assignment can break their congruences, so that operation is not used as a witness-restoration claim. The signed theorem identifies the main indices directly. If an entirely natural parent witness is desired, its auxiliary frame can then be reconstructed using the inherited simultaneous-completeness theorem.

7 Integer domains, sign certificates, and exact projection

The raw graph has 655 coordinates. The mixed-domain compiler retains nonnegativity exactly where the arithmetic proof requires it. The auxiliary U, S_i, R_i, divisibility quotients, and their product wires can remain signed. Main Pell pairs, all needed source and radix parameters, and all genuine remainder/inequality slacks retain their required bounds. The domain certificate checks this partition explicitly.

7.1 Which quotient signs are unnecessary

For an exact power, recurrence gives the identity

$$X_k + (n - A)Y_k - n^k = T_n \sum_{j=0}^{k-1} n^{k-1-j} Y_j.$$

An arbitrary integer quotient therefore suffices for the congruence proof. After the bounded residue is identified with n^k, this quotient is the displayed nonnegative sum. The eleven modular extraction quotients are also

determined by a positive number and its least nonnegative residue. The common root quotient is positive because $A \geq q+1$ and $2A(q+1) - (q+1)^2 - 1 > 0$.

Three further restorations are elementary. The initial endpoint has $x = (M_0+2)+(y+1)u$ with $0 \leq M_0+2 < y+1$ and $x \geq 0$, forcing $u \geq 0$. The product graph has $N = B+Du$, $0 \leq B < D$, $N \geq 0$, forcing its quotient nonnegative. Finally $F = Bv$, $F \geq 0$, $B \geq 1$ forces $v \geq 0$. These conclusions do not presume the product or binomial semantics before they have been established.

7.2 Encoding only the necessary signs

For an integer v,

$$v \geq 0 \quad \longleftrightarrow \quad \exists a, b, c \in \mathbb{Z} : \; v = a^2 + b^2 + c(c+1). \tag{13}$$

The right side is nonnegative. Conversely, apply the three-square theorem to $4v+1$, which is never of the excluded form $4^r(8s+7)$. Exactly one square is odd modulo four; rearranging yields (13). Each retained sign guard therefore costs three integer witnesses and one quadratic row.

The final certificate needs 96 guards. Exact implications from existing rows propagate the other required signs. They include positive monomial sums, affine squares, and the inherited implication $xA = B$ with a certified $A > 0$ and an appropriate lower bound on B. The checker validates every antecedent before admitting its conclusion; no omitted quotient sign is used to justify a retained semantic sign. The full finite ledger and its proof rules are included with each integer equation.

7.3 Projection and the single quartic

A monic defining equation $v - f = 0$ over $\mathbb{Z}$ allows exact elimination of v. Substitution in the remaining rows and reverse restoration $v = f$ prove existential equivalence. Over $\mathbb{N}$, an elimination must additionally certify that its restoration is nonnegative; the parent contractor uses positive monomial and rational sum-of-squares certificates for that purpose. The independent verifier checks every substitution as a polynomial identity and checks that residual degree never exceeds two.

The natural contraction deliberately retains the common sum W. Eliminating that one variable gives a 394-variable alternative but greatly increases the expanded term count. The selected 395-variable form has only 7,323 terms. The integer alternatives use two support thresholds: one reduces variables, and the other preserves smaller coefficients. These thresholds control expression growth; they are not mathematical assumptions.

Finally, over either domain a finite sum of integer squares is zero exactly when each residual is zero. This yields the literal quartics of Theorem 1.1. Their exact degree and coefficient counts are obtained by collecting all products in the squared sums, not by estimating the size of the compiler.

8 The internal-ZFC statement and its proof boundary

The desired statement is a provable equivalence, not a proof that the polynomial has no zeros. Combining the reductions gives

$$\exists \text{ polynomial zero} \;\longleftrightarrow\; \exists \text{ bounded arithmetic certificate} \;\longleftrightarrow\; C_{222}(1) \;\longleftrightarrow\; \exists p \, \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot).$$

Negating this equivalence proves the required no-zero/consistency equivalence in ZFC.

The new arguments are uniform integer identities, induction on finite Pell recurrences, Euclidean division, CRT, and finite certificate transformations. They can be carried out over ZFC's own ω and its definable integers. In particular they are not claims proved only for each externally standard numeral; the quantified proofs themselves are formalizable in the theory. The same uniform interpretation is required of the inherited calculus and arithmetic calibration.

We distinguish three kinds of evidence. The ordinary mathematical proof supplies the unbounded equivalences, with the inherited dependencies identified. Exact executable checkers establish the finite coefficient identities, domains, restorations, degree, and literal output binding. Small finite arithmetic fixtures catch transcription errors in the lemmas. Neither kind of computation replaces the ordinary proof. This package emits no new kernel-checked end-to-end theorem and no encoded object-language ZFC proof term. That formalization is a separate project task; it is not silently claimed complete.

9 Two additional twelve-integer improvements

The direct quartics optimize a different region from the few-variable constructions. To advance both regions, the archive also supplies two centered polynomials in twelve unrestricted integers, using the literal parent in the valuation-and-signed-certificates archive [6].

Construction	Integer variables	Exact degree	Live operations
Incoming rational form	12	2400	215
Incoming integral form	12	2912	209
Centered integral form	12	**2400**	**210**
Centered rational form	12	**2368**	216

These are exact polynomial translations of an unrestricted coordinate: $n_{\mathrm{old}} = n + QV$ or $n_{\mathrm{old}} = n + V$. The expressions Q, V do not depend on n, so the inverse translations are also integral and polynomial. Every zero is preserved in both directions, with no new positivity or parity hypothesis. The useful cancellation identities are

$$(Q-1)(SV+z)^2 - S^2(QV^2-1) = (Q-1)z^2 + 2(Q-1)SVz - S^2(V^2-1),$$
$$(SQV+z)^2 - S^2(Q-1)(QV^2-1) = z^2 + 2SQVz + S^2(QV^2+Q-1).$$

The integral implementation shares QV and computes $QV^2 = (QV)V$. The exact weighted-degree calculation, nonzero leading coefficients, complete circuit identities, inherited coefficient recipe, and standalone checkers are supplied under `supplemental/integer12/`.

These are incremental improvements, not the central new encoding idea. Their source architecture has different large fixed coefficients; the moderate-height claim for the direct quartics does not apply to them. Operation counts refer to live binary addition, subtraction, and multiplication nodes in a shared arithmetic circuit. They are neither expanded monomial counts nor binary coefficient lengths.

10 Reproduction, inspection, and remaining research

The archive includes the LaTeX source, compiled PDF, complete residual and expanded coefficient files, deterministic builders, independent checkers, mathematical reviews, corruption controls, source pins, and the original direct-construction dependency archive with its notices. Python 3.10 or later and the standard library suffice for the arithmetic reproduction. A conventional LaTeX installation compiles the report.

```
python3 verify.py
python3 -O verify.py
python3 reproduce.py --output /path/to/new-empty-directory
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The verifier treats delivered equation files as inputs and reconstructs coefficient collections independently. It rejects altered sign proofs, invalid restoration identities, and unauthorized signed roles. Reproduction builds the new source in an isolated directory from the pinned parent plus the new modules. No program searches for a refutation or a zero of the final polynomials.

The new route suggests a broader optimization principle: when a Diophantine certificate consumes only a residue of an expensive arithmetic value, represent that residue directly and prove simultaneous completeness for the shared parameters. Exact values are needed only where their magnitude or arithmetic meaning is actually consumed. Here the extra freedom in the Pell base is what makes this principle an exact construction instead of an unsupported identification of independently chosen witnesses.

Further reductions may exploit additional relationships among requested indices, fewer locally bounded proof parameters, or stronger sign inference. Such changes must preserve their complete domains and cofinal witness construction. None is assumed in the delivered result. Global optimality, an ultimate few-variable record, and a fresh end-to-end formal proof certificate remain outside what has been proved.

References

[1] V. Reshetnikov and ProveIt contributors, *DiophantineConsistency research*, comparison snapshot `ec80841c07e91be8956ce6ed4af9a9c0b782d3d9`. https://github.com/VladimirReshetnikov/ProveIt/tree/ec80841c07e91be8956ce6ed4af9a9c0b782d3d9/Computability/DiophantineConsistency.

[2] ProveIt incoming contribution, *Residue certificates and nonnegative gates for the consistency of ZFC*, `ConZFC_407_888_Quartics_2026-10-24.zip`, same snapshot. https://github.com/VladimirReshetnikov/ProveIt/blob/ec80841c07e91be8956ce6ed4af9a9c0b782d3d9/docs/incoming/ConZFC_407_888_Quartics_2026-10-24.zip. The complete archive is included in `inputs/`.

[3] ProveIt incoming contribution, `ConZFC_Shared_Pell_1873.zip`, especially `proofs/SIGNED_INDEX_THEOREM.md` and its independent review. https://github.com/VladimirReshetnikov/ProveIt/blob/ec80841c07e91be8956ce6ed4af9a9c0b782d3d9/docs/incoming/ConZFC_Shared_Pell_1873.zip.

[4] ProveIt incoming contribution, `ConZFC_Direct_Quartics_3776_3671_2026-10-23.zip`, especially the sign-orbit and guard-minimization proofs. https://github.com/VladimirReshetnikov/ProveIt/blob/ec80841c07e91be8956ce6ed4af9a9c0b782d3d9/docs/incoming/ConZFC_Direct_Quartics_3776_3671_2026-10-23.zip.

[5] ProveIt incoming contribution, `ZFC_Diophantine_Joint_Guards_2026-10-24.zip`, same snapshot. https://github.com/VladimirReshetnikov/ProveIt/blob/ec80841c07e91be8956ce6ed4af9a9c0b782d3d9/docs/incoming/ZFC_Diophantine_Joint_Guards_2026-10-24.zip.

[6] ProveIt incoming contribution, `ConZFC_Valuation_and_Signed_Certificates_2026-10-23.zip`, especially the literal `integer12` subtree. https://github.com/VladimirReshetnikov/ProveIt/blob/ec80841c07e91be8956ce6ed4af9a9c0b782d3d9/docs/incoming/ConZFC_Valuation_and_Signed_Certificates_2026-10-23.zip.

[7] Mathlib contributors, *Pell's equation and Matiyasevic's theorem*, especially `eq_pell`, `dvd_of_ysq_dvd`, `xn_modEq_x2n_add`, `modEq_of_xn_modEq`, and `matiyasevic`. https://leanprover-community.github.io/mathlib4_docs/Mathlib/NumberTheory/PellMatiyasevic.html.

[8] M. Carneiro, *A Lean formalization of Matiyasevič's theorem*, 2018, arXiv:1802.01795. https://arxiv.org/abs/1802.01795.

A The complete 395-variable natural equation

Each x_i independently ranges over $\mathbb{N}$, including zero. The exact single polynomial is

$$\boxed{P_{395}(x_0, \ldots, x_{394}) = \sum_{j=0}^{320} R_j(x_0, \ldots, x_{394})^2 = 0.}$$

The following list contains every residual and every coefficient. The machine-readable order is `equations/natural395/equation.json`; the fully collected expansion is in its sibling `expanded_polynomial.json`. Auxiliary names used in the construction do not add coordinates beyond the numbered variables in this list.

Residual	Polynomial (set each to zero)
R_0	$-x_3^2 + x_4$
R_1	$-x_4^2 + x_5$
R_2	$-x_5^2 + x_6$
R_3	$-x_6^2 + x_7$
R_4	$-x_7^2 + x_8$
R_5	$-x_8^2 + x_9$
R_6	$-x_9^2 + x_{10}$
R_7	$-x_{10}^2 + x_{11}$
R_8	$-x_{11}^2 + x_{12}$
R_9	$-x_{12}^2 + x_{13}$
R_{10}	$-x_{13}^2 + x_{14}$
R_{11}	$-x_{14}^2 + x_{15}$
R_{12}	$-x_{15}^2 + x_{16}$
R_{13}	$-x_{16}^2 + x_{17}$
R_{14}	$-x_{17}^2 + x_{18}$
R_{15}	$-x_{18}^2 + x_{19}$
R_{16}	$-x_{19}^2 + x_{20}$
R_{17}	$-x_{20}^2 + x_{21}$
R_{18}	$-x_{21}^2 + x_{22}$
R_{19}	$-x_{22}^2 + x_{23}$
R_{20}	$-x_{23}^2 + x_{24}$
R_{21}	$-1 - x_2 + x_{24}^2$
R_{22}	$-4 - 2x_0 - x_0x_2 - x_0x_{26} - 2x_2 - 2x_{26} + x_{28}$
R_{23}	$-2 + x_1 - x_2 - x_2x_{25} - 3x_{25} - x_{25}x_{26}$
R_{24}	$-1 + x_1 - x_{27} - x_{27}x_{28}$
R_{25}	$-25 - 10x_0 - x_0^2 - 2x_0x_1 - 4x_0x_2 - 2x_0x_{26} - 10x_1 - x_1^2 - 4x_1x_2 - 2x_1x_{26} - 20x_2 - 4x_2^2 - 4x_2x_{26}$ $- 10x_{26} - x_{26}^2 + x_{38}$
R_{26}	$-x_{38}^2 + x_{39}$
R_{27}	$-x_{39}^2 + x_{40}$
R_{28}	$-x_{40}^2 + x_{41}$
R_{29}	$-x_{41}^2 + x_{42}$
R_{30}	$-x_{42}^2 + x_{43}$
R_{31}	$-x_{43}^2 + x_{44}$
R_{32}	$-x_{44}^2 + x_{45}$
R_{33}	$-x_{45}^2 + x_{46}$
R_{34}	$-x_{46}^2 + x_{47}$
R_{35}	$-x_{47}^2 + x_{48}$
R_{36}	$-x_{48}^2 + x_{49}$
R_{37}	$-x_{49}^2 + x_{50}$
R_{38}	$-x_{50}^2 + x_{51}$

Residual	Polynomial (continued)
R_{39}	$-x_{51}^2+x_{52}$
R_{40}	$-x_{52}^2+x_{53}$
R_{41}	$-x_{53}^2+x_{54}$
R_{42}	$-x_{54}^2+x_{55}$
R_{43}	$-x_{55}^2+x_{56}$
R_{44}	$-x_{56}^2+x_{57}$
R_{45}	$-x_{57}^2+x_{58}$
R_{46}	$-x_{58}^2+x_{59}$
R_{47}	$-x_{59}^2+x_{60}$
R_{48}	$-x_{61}-x_{61}x_{64}-x_{61}x_{65}+x_{63}+x_{65}$
R_{49}	$-x_0x_{61}-2x_{61}+x_{66}$
R_{50}	$-x_{69}-x_{69}x_{72}-x_{69}x_{73}+x_{71}+x_{73}$
R_{51}	$-x_{68}x_{69}+x_{75}$
R_{52}	$2-2x_{61}-x_{61}x_{67}+x_{76}+x_{78}$
R_{53}	$x_{74}x_{75}-x_{76}-x_{76}x_{77}-x_{77}-x_{77}x_{78}$
R_{54}	$1+x_{37}x_{61}+x_{61}-x_{76}$
R_{55}	$-6-5x_{29}-x_{29}^2-2x_{29}x_{30}-5x_{30}-x_{30}^2+x_{79}$
R_{56}	$-72-68x_{30}-16x_{30}^2-16x_{30}x_{79}-34x_{79}-4x_{79}^2+x_{80}$
R_{57}	$-30-11x_{29}-x_{29}^2-4x_{29}x_{80}-22x_{80}-4x_{80}^2+x_{81}$
R_{58}	$-380-312x_{80}-64x_{80}^2-32x_{80}x_{81}-78x_{81}-4x_{81}^2+x_{82}$
R_{59}	$-6-5x_{30}-x_{30}^2-2x_{30}x_{31}-5x_{31}-x_{31}^2+x_{83}$
R_{60}	$-6-5x_{29}-x_{29}^2-2x_{29}x_{31}-5x_{31}-x_{31}^2+x_{84}$
R_{61}	$-414-22x_{29}-2x_{29}^2-16x_{29}x_{31}-8x_{29}x_{83}-200x_{30}-32x_{30}^2-64x_{30}x_{31}-32x_{30}x_{79}-32x_{30}x_{84}-320x_{31}$ $-64x_{31}^2-32x_{31}x_{79}-32x_{31}x_{83}-32x_{31}x_{84}-100x_{79}-8x_{79}^2-16x_{79}x_{84}-52x_{83}-8x_{83}^2-108x_{84}-8x_{84}^2$ $+x_{85}$
R_{62}	$-1344-800x_{30}-128x_{30}^2-256x_{30}x_{31}-128x_{30}x_{79}-128x_{30}x_{84}-864x_{31}-128x_{31}^2-128x_{31}x_{79}$ $-128x_{31}x_{84}-400x_{79}-32x_{79}^2-64x_{79}x_{84}-432x_{84}-32x_{84}^2-2x_{85}-2x_{85}^2+x_{86}$
R_{63}	$-247-168x_{29}-32x_{29}^2-64x_{29}x_{30}-32x_{29}x_{83}-32x_{29}x_{84}-188x_{30}-32x_{30}^2-32x_{30}x_{83}-32x_{30}x_{84}$ $-2x_{79}-92x_{83}-8x_{83}^2-16x_{83}x_{84}-84x_{84}-8x_{84}^2+x_{87}$
R_{64}	$-960-672x_{29}-128x_{29}^2-256x_{29}x_{30}-128x_{29}x_{83}-128x_{29}x_{84}-736x_{30}-128x_{30}^2-128x_{30}x_{83}$ $-128x_{30}x_{84}-368x_{83}-32x_{83}^2-64x_{83}x_{84}-336x_{84}-32x_{84}^2-2x_{87}-2x_{87}^2+x_{88}$
R_{65}	$-276-30x_{29}-2x_{29}^2-16x_{29}x_{31}-8x_{29}x_{83}-30x_{30}-2x_{30}^2-16x_{30}x_{31}-8x_{30}x_{84}-272x_{31}-64x_{31}^2$ $-32x_{31}x_{83}-32x_{31}x_{84}-68x_{83}-8x_{83}^2-68x_{84}-8x_{84}^2+x_{89}$
R_{66}	$-552-120x_{30}-8x_{30}^2-64x_{30}x_{31}-32x_{30}x_{84}-544x_{31}-128x_{31}^2-128x_{31}x_{84}-272x_{84}-32x_{84}^2-2x_{89}$ $-2x_{89}^2+x_{90}$
R_{67}	$-247-4x_{29}-352x_{30}-128x_{30}^2-64x_{30}x_{79}-64x_{30}x_{83}-92x_{79}-8x_{79}^2-16x_{79}x_{83}-84x_{83}-8x_{83}^2-2x_{84}$ $+x_{91}$
R_{68}	$-960-1408x_{30}-512x_{30}^2-256x_{30}x_{79}-256x_{30}x_{83}-368x_{79}-32x_{79}^2-64x_{79}x_{83}-336x_{83}-32x_{83}^2$ $-2x_{91}-2x_{91}^2+x_{92}$
R_{69}	$-56-60x_{30}-16x_{30}^2-16x_{30}x_{79}-30x_{79}-4x_{79}^2+x_{93}$
R_{70}	$-30-11x_{29}-x_{29}^2-4x_{29}x_{93}-22x_{93}-4x_{93}^2+x_{94}$
R_{71}	$-380-312x_{93}-64x_{93}^2-32x_{93}x_{94}-78x_{94}-4x_{94}^2+x_{95}$
R_{72}	$-30-11x_{30}-x_{30}^2-4x_{30}x_{95}-22x_{95}-4x_{95}^2+x_{96}$
R_{73}	$-349-420x_{30}-128x_{30}^2-128x_{30}x_{79}-208x_{79}-32x_{79}^2-2x_{83}+x_{97}$
R_{74}	$-1368-1664x_{30}-512x_{30}^2-512x_{30}x_{79}-832x_{79}-128x_{79}^2-2x_{97}-2x_{97}^2+x_{98}$
R_{75}	$-56-60x_{29}-16x_{29}^2-16x_{29}x_{84}-8x_{29}x_{98}-30x_{84}-4x_{84}^2-4x_{84}x_{98}-15x_{98}-x_{98}^2+x_{99}$
R_{76}	$-6-5x_{31}-x_{31}^2-2x_{31}x_{32}-5x_{32}-x_{32}^2+x_{100}$
R_{77}	$-224-26x_{29}-2x_{29}^2-16x_{29}x_{32}-8x_{29}x_{100}-288x_{31}-128x_{31}^2-64x_{31}x_{83}-64x_{31}x_{84}-88x_{32}-32x_{32}^2$ $-32x_{32}x_{100}-76x_{83}-8x_{83}^2-16x_{83}x_{84}-68x_{84}-8x_{84}^2-44x_{100}-8x_{100}^2+x_{101}$
R_{78}	$-640-1152x_{31}-512x_{31}^2-256x_{31}x_{83}-256x_{31}x_{84}-304x_{83}-32x_{83}^2-64x_{83}x_{84}-272x_{84}-32x_{84}^2$ $-2x_{101}-2x_{101}^2+x_{102}$
R_{79}	$-241-168x_{29}-32x_{29}^2-64x_{29}x_{30}-32x_{29}x_{83}-32x_{29}x_{84}-184x_{30}-32x_{30}^2-32x_{30}x_{83}-32x_{30}x_{84}-x_{31}$ $-92x_{83}-8x_{83}^2-16x_{83}x_{84}-84x_{84}-8x_{84}^2+x_{103}$

Residual	Polynomial (continued)
R_{80}	$-42-52x_{31}-16x_{31}^2-16x_{31}x_{83}-26x_{83}-4x_{83}^2+x_{104}$
R_{81}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{104}-22x_{104}-4x_{104}^2+x_{105}$
R_{82}	$-306-35x_{30}-x_{30}^2-16x_{30}x_{104}-4x_{30}x_{105}-280x_{104}-64x_{104}^2-32x_{104}x_{105}-70x_{105}-4x_{105}^2+x_{106}$
R_{83}	$-4556-4320x_{104}-1024x_{104}^2-512x_{104}x_{105}-128x_{104}x_{106}-1080x_{105}-64x_{105}^2-32x_{105}x_{106}-270x_{106}-4x_{106}^2+x_{107}$
R_{84}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{107}-22x_{107}-4x_{107}^2+x_{108}$
R_{85}	$-306-35x_{30}-x_{30}^2-16x_{30}x_{107}-4x_{30}x_{108}-280x_{107}-64x_{107}^2-32x_{107}x_{108}-70x_{108}-4x_{108}^2+x_{109}$
R_{86}	$-1028-672x_{29}-128x_{29}^2-256x_{29}x_{30}-128x_{29}x_{83}-128x_{29}x_{84}-736x_{30}-128x_{30}^2-128x_{30}x_{83}-128x_{30}x_{84}-368x_{83}-32x_{83}^2-64x_{83}x_{84}-336x_{84}-32x_{84}^2-2x_{103}-2x_{103}^2-32x_{107}-8x_{108}-2x_{109}+x_{110}$
R_{87}	$-264-128x_{107}-32x_{108}-8x_{109}-2x_{110}-2x_{110}^2+x_{111}$
R_{88}	$-2-3x_{29}-x_{29}^2-2x_{29}x_{111}-3x_{111}-x_{111}^2+x_{112}$
R_{89}	$-12-28x_{111}-16x_{111}^2-16x_{111}x_{112}-14x_{112}-4x_{112}^2+x_{113}$
R_{90}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{113}-22x_{113}-4x_{113}^2+x_{114}$
R_{91}	$-306-35x_{29}-x_{29}^2-16x_{29}x_{113}-4x_{29}x_{114}-280x_{113}-64x_{113}^2-32x_{113}x_{114}-70x_{114}-4x_{114}^2+x_{115}$
R_{92}	$-4556-4320x_{113}-1024x_{113}^2-512x_{113}x_{114}-128x_{113}x_{115}-1080x_{114}-64x_{114}^2-32x_{114}x_{115}-270x_{115}-4x_{115}^2+x_{116}$
R_{93}	$-90-76x_{31}-16x_{31}^2-16x_{31}x_{83}-16x_{31}x_{116}-38x_{83}-4x_{83}^2-8x_{83}x_{116}-38x_{116}-4x_{116}^2+x_{117}$
R_{94}	$-4556-4320x_{107}-1024x_{107}^2-512x_{107}x_{108}-128x_{107}x_{109}-1080x_{108}-64x_{108}^2-32x_{108}x_{109}-270x_{109}-4x_{109}^2+x_{118}$
R_{95}	$-682-70x_{30}-2x_{30}^2-32x_{30}x_{116}-8x_{30}x_{117}-592x_{116}-128x_{116}^2-64x_{116}x_{117}-148x_{117}-8x_{117}^2-2x_{118}+x_{119}$
R_{96}	$-x_{119}-x_{119}^2+x_{120}$
R_{97}	$-42-52x_{30}-16x_{30}^2-8x_{30}x_{31}-16x_{30}x_{79}-13x_{31}-x_{31}^2-4x_{31}x_{79}-26x_{79}-4x_{79}^2+x_{121}$
R_{98}	$-2278-2264x_{30}-800x_{30}^2-800x_{30}x_{79}-160x_{30}x_{121}-70x_{31}-2x_{31}^2-32x_{31}x_{118}-8x_{31}x_{120}-1132x_{79}-200x_{79}^2-80x_{79}x_{121}-592x_{118}-128x_{118}^2-64x_{118}x_{120}-148x_{120}-8x_{120}^2-228x_{121}-8x_{121}^2+x_{122}$
R_{99}	$-2712-280x_{31}-8x_{31}^2-128x_{31}x_{118}-32x_{31}x_{120}-2368x_{118}-512x_{118}^2-256x_{118}x_{120}-592x_{120}-32x_{120}^2-2x_{122}-2x_{122}^2+x_{123}$
R_{100}	$-2-3x_{123}-x_{123}^2+x_{124}$
R_{101}	$-1-x_2-x_2x_{34}-x_{34}+x_{125}$
R_{102}	$-x_{33}^2+2x_{33}x_{34}-x_{34}^2+x_{126}$
R_{103}	$-2x_{82}x_{125}-4x_{125}+x_{127}$
R_{104}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{127}-x_{34}^2-2x_{34}x_{127}-x_{127}^2+x_{128}$
R_{105}	$-x_{86}x_{125}+x_{129}$
R_{106}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{129}-x_{34}^2-2x_{34}x_{129}-x_{129}^2+x_{130}$
R_{107}	$-x_{88}x_{125}+x_{131}$
R_{108}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{131}-x_{34}^2-2x_{34}x_{131}-x_{131}^2+x_{132}$
R_{109}	$-x_{90}x_{125}+x_{133}$
R_{110}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{133}-x_{34}^2-2x_{34}x_{133}-x_{133}^2+x_{134}$
R_{111}	$-x_{92}x_{125}+x_{135}$
R_{112}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{135}-x_{34}^2-2x_{34}x_{135}-x_{135}^2+x_{136}$
R_{113}	$-4x_{30}x_{125}-2x_{96}x_{125}-4x_{125}+x_{137}$
R_{114}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{137}-x_{34}^2-2x_{34}x_{137}-x_{137}^2+x_{138}$
R_{115}	$-4x_{98}x_{125}-2x_{99}x_{125}+x_{139}$
R_{116}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{139}-x_{34}^2-2x_{34}x_{139}-x_{139}^2+x_{140}$
R_{117}	$-x_{102}x_{125}+x_{141}$
R_{118}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{141}-x_{34}^2-2x_{34}x_{141}-x_{141}^2+x_{142}$
R_{119}	$-x_{29}^2+2x_{29}x_{31}-x_{31}^2+x_{143}$
R_{120}	$-x_{30}^2+2x_{30}x_{31}-x_{31}^2+x_{144}$
R_{121}	$-x_{143}x_{144}+x_{145}$
R_{122}	$-4-4x_{32}-x_{32}^2+x_{146}$
R_{123}	$-x_{145}^2+2x_{145}x_{146}-x_{146}^2+x_{147}$

Residual	Polynomial (continued)
R_{124}	$-2x_{124}x_{125} - 4x_{125} + x_{148}$
R_{125}	$-x_{33}^2 + 2x_{33}x_{34} + 2x_{33}x_{148} - x_{34}^2 - 2x_{34}x_{148} - x_{148}^2 + x_{149}$
R_{126}	$-1 - x_2 - x_2x_{30} - x_{30} + x_{150}$
R_{127}	$-1 - x_{34} - x_{34}x_{150} - x_{150} + x_{151}$
R_{128}	$-1 - x_2 - x_2x_{29} - x_{29} + x_{152}$
R_{129}	$-4 - 4x_2 - 4x_2x_{30} - 2x_2x_{79} - 4x_{30} - 2x_{79} + x_{153}$
R_{130}	$-1 - x_{33} - x_{33}x_{152} - x_{152} + x_{154}$
R_{131}	$-x_{153}x_{154} - x_{154} + x_{155}$
R_{132}	$-x_{151}^2 + 2x_{151}x_{155} - x_{155}^2 + x_{156}$
R_{133}	$-6 - 6x_2 - 4x_2x_{30} - 2x_2x_{79} - 4x_{30} - 2x_{79} + x_{157}$
R_{134}	$-1 - x_{34} - x_{34}x_{157} - x_{157} + x_{158}$
R_{135}	$-1 - x_{33} - x_{33}x_{150} - x_{150} + x_{159}$
R_{136}	$-x_{158}^2 + 2x_{158}x_{159} - x_{159}^2 + x_{160}$
R_{137}	$-x_{126}x_{128} + x_{161}$
R_{138}	$-x_{130}x_{161} + x_{162}$
R_{139}	$-x_{132}x_{162} + x_{163}$
R_{140}	$-x_{134}x_{163} + x_{164}$
R_{141}	$-x_{136}x_{164} + x_{165}$
R_{142}	$-x_{138}x_{165} + x_{166}$
R_{143}	$-x_{140}x_{166} + x_{167}$
R_{144}	$-x_{142}x_{167} + x_{168}$
R_{145}	$-x_{147}x_{168} - x_{149}x_{168} + x_{169}$
R_{146}	$-x_{156}x_{169} + x_{170}$
R_{147}	$-2 - x_2 - x_2x_{37} - x_{26} - x_{26}x_{37} - 2x_{37} + x_{171}$
R_{148}	$-1 - x_{35} - x_{35}x_{171} - x_{171} + x_{172}$
R_{149}	$-3 - x_2 - x_2x_{36} - x_{26} - x_{26}x_{36} - 3x_{36} - x_{36}x_{171} - x_{171} + x_{173}$
R_{150}	$2 - 4x_1 + 2x_1^2 - 2x_1x_{33} - 2x_1x_{34} - 2x_1x_{172} - 2x_1x_{173} + 2x_{33} + x_{33}^2 + 2x_{33}x_{172} + 2x_{34} + x_{34}^2 + 2x_{34}x_{173} - x_{76}x_{174} + x_{160}x_{170} + 2x_{172} + x_{172}^2 + 2x_{173} + x_{173}^2$
R_{151}	$-8 - x_{29} - x_{30} - x_{31} - x_{32} - x_{33} - x_{34} - x_{35} - x_{36} + 8x_{76} - x_{175}$
R_{152}	$2 - x_{63}x_{76} + x_{64} + x_{65} - x_{76} + x_{263}$
R_{153}	$-2 - x_{64} - x_{64}x_{261} - x_{64}x_{262} - x_{65} - x_{65}x_{261} - x_{65}x_{262} + x_{176} - 2x_{261} - x_{261}x_{263} - 2x_{262} - x_{262}x_{263} - x_{263}$
R_{154}	$-x_{177} - x_{177}x_{180} + x_{181}$
R_{155}	$-x_{181}^2 + x_{182}$
R_{156}	$-1 - 2x_{177} - x_{177}^2 + x_{183}$
R_{157}	$-1 + x_{178}^2 + x_{182} - x_{182}x_{183}$
R_{158}	$-10 - 6x_{64} - x_{64}^2 - 2x_{64}x_{65} + 2x_{64}x_{178} + 2x_{64}x_{179} - 2x_{64}x_{263} - 6x_{65} - x_{65}^2 + 2x_{65}x_{178} + 2x_{65}x_{179} - 2x_{65}x_{263} - x_{176}x_{184} + 6x_{178} + 2x_{178}x_{263} + 6x_{179} + 2x_{179}x_{263} - 6x_{263} - x_{263}^2$
R_{159}	$-x_{186}^2 + x_{189}$
R_{160}	$1 - x_{178}^2 - 2x_{178}x_{179} - x_{179}^2 + x_{211}$
R_{161}	$-1 + x_{185}^2 - x_{189}x_{211}$
R_{162}	$-4x_{186} - 4x_{186}x_{187} + x_{190}$
R_{163}	$1 - x_{178} - x_{179} - x_{185}x_{188} + x_{190}$
R_{164}	$-1 - 2x_0 - x_0^2 - 2x_0x_{194} - 2x_{194} - x_{194}^2 + x_{195}$
R_{165}	$-1 + x_{191}^2 - x_{195}x_{211}$
R_{166}	$-x_{193}^2 + x_{196}$
R_{167}	$-1 - 2x_{190} - x_{190}^2 + x_{197}$
R_{168}	$-1 + x_{192}^2 + x_{196} - x_{196}x_{197}$
R_{169}	$-x_{185}x_{198} - x_{191} + x_{192}$
R_{170}	$-1 - x_0 - 4x_0x_{199} + x_{193} - 4x_{194}x_{199} - 4x_{199}$
R_{171}	$-5 - 4x_0 - x_0^2 + 2x_0x_{178} + 2x_0x_{179} + 4x_{178} + 4x_{179} - x_{200}$
R_{172}	$-x_0x_{178} - x_0x_{179} - x_{178} - x_{178}x_{194} - x_{179} - x_{179}x_{194} + x_{202}$
R_{173}	$2 + 3x_0 + x_0^2 + x_0x_{194} - x_3 + x_{191} + 2x_{194} - x_{200}x_{201} - x_{202}$

Residual	Polynomial (continued)
R_{174}	$-x_{60}^2 - 2x_{60}x_{206} - x_{206}^2 + x_{207}$
R_{175}	$-1 + x_{203}^2 - x_{207}x_{211}$
R_{176}	$-x_{205}^2 + x_{208}$
R_{177}	$-1 - x_{197}x_{208} + x_{204}^2 + x_{208}$
R_{178}	$-x_{185}x_{209} - x_{203} + x_{204}$
R_{179}	$-x_{60} - 4x_{60}x_{210} + x_{205} - 4x_{206}x_{210}$
R_{180}	$-2x_{60}x_{178} - 2x_{60}x_{179} - 2x_{178}x_{206} - 2x_{179}x_{206} + x_{212}$
R_{181}	$-2x_{178}x_{203} - 2x_{179}x_{203} + x_{213}$
R_{182}	$-x_{60} - 2x_{60}x_{211} - x_{206} - 2x_{206}x_{211} + x_{214}$
R_{183}	$-5 - 4x_{60} - x_{60}^2 + 2x_{60}x_{178} + 2x_{60}x_{179} + 4x_{178} + 4x_{179} - x_{215}$
R_{184}	$x_{60}x_{213} + x_{60}x_{214} - x_{62} - x_{178}x_{213} - x_{178}x_{214} - x_{179}x_{213} - x_{179}x_{214} + x_{203} + 2x_{203}x_{211} + x_{211}x_{212} + 2x_{213} + 2x_{214} - x_{215}x_{216}$
R_{185}	$-2 - 2x_{62} - x_{62}^2 + 2x_{62}x_{178} + 2x_{62}x_{179} + 2x_{178} + 2x_{179} - x_{217}$
R_{186}	$x_{60} + x_{60}x_{62} - x_{60}x_{178} - x_{60}x_{179} + x_{62}x_{206} - x_{63} - x_{178}x_{206} - x_{179}x_{206} + x_{203} + x_{206} - x_{217}x_{218}$
R_{187}	$-2 - 2x_{66} - x_{66}^2 + 2x_{66}x_{178} + 2x_{66}x_{179} + 2x_{178} + 2x_{179} - x_{219}$
R_{188}	$1 + x_0 + x_0x_{66} + x_{66} + x_{66}x_{194} - x_{67} + x_{191} + x_{194} - x_{202} - x_{219}x_{220}$
R_{189}	$-1 - x_{61}^2 + 2x_{61}x_{178} + 2x_{61}x_{179} - x_{221}$
R_{190}	$x_0x_{61} + x_{61} + x_{61}x_{194} - x_{68} + x_{191} - x_{202} - x_{221}x_{222}$
R_{191}	$-2x_0x_{178} - 2x_0x_{179} - 2x_{178} - 2x_{178}x_{194} - 2x_{179} - 2x_{179}x_{194} + x_{223}$
R_{192}	$-2x_{178}x_{191} - 2x_{179}x_{191} + x_{224}$
R_{193}	$-1 - x_0 - 2x_0x_{211} - x_{194} - 2x_{194}x_{211} - 2x_{211} + x_{225}$
R_{194}	$-10 - 6x_0 - x_0^2 + 2x_0x_{178} + 2x_0x_{179} + 6x_{178} + 6x_{179} - x_{226}$
R_{195}	$x_0x_{224} + x_0x_{225} - x_{70} - x_{178}x_{224} - x_{178}x_{225} - x_{179}x_{224} - x_{179}x_{225} + x_{191} + 2x_{191}x_{211} + x_{211}x_{223} + 3x_{224} + 3x_{225} - x_{226}x_{227}$
R_{196}	$-2 - 2x_{70} - x_{70}^2 + 2x_{70}x_{178} + 2x_{70}x_{179} + 2x_{178} + 2x_{179} - x_{228}$
R_{197}	$1 + x_0 + x_0x_{70} + x_{70} + x_{70}x_{194} - x_{71} + x_{191} + x_{194} - x_{202} - x_{228}x_{229}$
R_{198}	$-5 - 4x_{64} - x_{64}^2 - 2x_{64}x_{65} + 2x_{64}x_{178} + 2x_{64}x_{179} - 2x_{64}x_{263} - 4x_{65} - x_{65}^2 + 2x_{65}x_{178} + 2x_{65}x_{179} - 2x_{65}x_{263} + 4x_{178} + 2x_{178}x_{263} + 4x_{179} + 2x_{179}x_{263} - x_{230} - 4x_{263} - x_{263}^2$
R_{199}	$-1 + 2x_{60} + x_{60}x_{64} + x_{60}x_{65} - x_{60}x_{178} - x_{60}x_{179} + x_{60}x_{263} + x_{64}x_{206} + x_{65}x_{206} - x_{178}x_{206} - x_{179}x_{206} + x_{203} + 2x_{206} + x_{206}x_{263} - x_{230}x_{231} - x_{261} - x_{262}$
R_{200}	$1 + 2x_0 + x_0x_{64} + x_0x_{65} + x_0x_{263} + x_{64} + x_{64}x_{194} + x_{65} + x_{65}x_{194} + x_{191} + 2x_{194} + x_{194}x_{263} - x_{202} - x_{230}x_{232} + x_{263} - x_{273} - x_{274}$
R_{201}	$-1 + x_0x_{211} + x_{64}x_{191} + x_{64}x_{202} + x_{65}x_{191} + x_{65}x_{202} - x_{178}x_{202} - x_{179}x_{202} + 2x_{191} + x_{191}x_{263} + x_{194}x_{211} + 2x_{202} + x_{202}x_{263} + x_{211} - x_{230}x_{233} - x_{298} - x_{299}$
R_{202}	$-9 - 6x_2 - x_2^2 - 2x_2x_{26} - 2x_2x_{237} - 6x_{26} - x_{26}^2 - 2x_{26}x_{237} - 6x_{237} - x_{237}^2 + x_{238}$
R_{203}	$-1 - x_{211}x_{238} + x_{234}^2$
R_{204}	$-x_{236}^2 + x_{239}$
R_{205}	$-1 - x_{197}x_{239} + x_{235}^2 + x_{239}$
R_{206}	$-x_{185}x_{240} - x_{234} + x_{235}$
R_{207}	$-3 - x_2 - 4x_2x_{241} - x_{26} - 4x_{26}x_{241} + x_{236} - 4x_{237}x_{241} - 12x_{241}$
R_{208}	$5 + 2x_2 + x_2x_{64} + x_2x_{65} - x_2x_{178} - x_2x_{179} + x_2x_{263} + 2x_{26} + x_{26}x_{64} + x_{26}x_{65} - x_{26}x_{178} - x_{26}x_{179} + x_{26}x_{263} + 3x_{64} + x_{64}x_{237} + 3x_{65} + x_{65}x_{237} - 3x_{178} - x_{178}x_{237} - 3x_{179} - x_{179}x_{237} - x_{230}x_{242} + x_{234} + 2x_{237} + x_{237}x_{263} + 3x_{263} - x_{343} - x_{344}$
R_{209}	$-1 - 2x_1 - x_1^2 - 2x_1x_{246} - 2x_{246} - x_{246}^2 + x_{247}$
R_{210}	$-1 - x_{211}x_{247} + x_{243}^2$
R_{211}	$-x_{245}^2 + x_{248}$
R_{212}	$-1 - x_{197}x_{248} + x_{244}^2 + x_{248}$
R_{213}	$-x_{185}x_{249} - x_{243} + x_{244}$
R_{214}	$-1 - x_1 - 4x_1x_{250} + x_{245} - 4x_{246}x_{250} - 4x_{250}$
R_{215}	$1 + 2x_1 + x_1x_{64} + x_1x_{65} - x_1x_{178} - x_1x_{179} + x_1x_{263} + x_{64} + x_{64}x_{246} + x_{65} + x_{65}x_{246} - x_{178} - x_{178}x_{246} - x_{179} - x_{179}x_{246} - x_{230}x_{251} + x_{243} + 2x_{246} + x_{246}x_{263} + x_{263} - x_{366} - x_{367}$
R_{216}	$-x_{60}^2 - 2x_{60}x_{62} - 2x_{60}x_{255} - x_{62}^2 - 2x_{62}x_{255} - x_{255}^2 + x_{256}$
R_{217}	$-1 - x_{211}x_{256} + x_{252}^2$

Residual	Polynomial (continued)
R_{218}	$-x_{254}^2+x_{257}$
R_{219}	$-1-x_{197}x_{257}+x_{253}^2+x_{257}$
R_{220}	$-x_{185}x_{258}-x_{252}+x_{253}$
R_{221}	$-x_{60}-4x_{60}x_{259}-x_{62}-4x_{62}x_{259}+x_{254}-4x_{255}x_{259}$
R_{222}	$-1+3x_{60}+x_{60}x_{64}+x_{60}x_{65}-x_{60}x_{178}-x_{60}x_{179}+x_{60}x_{263}+3x_{62}+x_{62}x_{64}+x_{62}x_{65}-x_{62}x_{178}-x_{62}x_{179}$ $+x_{62}x_{263}-x_{64}+x_{64}x_{255}-x_{64}x_{261}-x_{64}x_{262}-x_{65}+x_{65}x_{255}-x_{65}x_{261}-x_{65}x_{262}-x_{176}x_{260}-x_{178}x_{255}$ $-x_{179}x_{255}+x_{252}+3x_{255}+x_{255}x_{263}-2x_{261}-x_{262}$
R_{223}	$-1-2x_0-x_0^2-2x_0x_{70}-2x_0x_{267}-2x_{70}-x_{70}^2-2x_{70}x_{267}-2x_{267}-x_{267}^2+x_{268}$
R_{224}	$-1-x_{211}x_{268}+x_{264}^2$
R_{225}	$-x_{266}^2+x_{269}$
R_{226}	$-1-x_{197}x_{269}+x_{265}^2+x_{269}$
R_{227}	$-x_{185}x_{270}-x_{264}+x_{265}$
R_{228}	$-1-x_0-4x_0x_{271}-x_{70}-4x_{70}x_{271}+x_{266}-4x_{267}x_{271}-4x_{271}$
R_{229}	$-2-x_{64}-x_{64}x_{273}-x_{64}x_{274}-x_{65}-x_{65}x_{273}-x_{65}x_{274}-x_{263}-x_{263}x_{273}-x_{263}x_{274}-2x_{273}-2x_{274}+x_{275}$
R_{230}	$x_{64}+x_{65}-x_{72}-x_{73}+x_{263}-x_{276}$
R_{231}	$2+3x_0+x_0x_{64}+x_0x_{65}-x_0x_{178}-x_0x_{179}+x_0x_{263}+x_{64}+x_{64}x_{70}+x_{64}x_{267}+x_{65}+x_{65}x_{70}+x_{65}x_{267}$ $+3x_{70}-x_{70}x_{178}-x_{70}x_{179}+x_{70}x_{263}-x_{72}-x_{72}x_{273}-x_{72}x_{274}-x_{73}-x_{73}x_{273}-x_{73}x_{274}-x_{178}-x_{178}x_{267}$ $-x_{179}-x_{179}x_{267}+x_{263}+x_{263}x_{267}+x_{264}+3x_{267}-x_{272}x_{275}-2x_{273}-x_{274}$
R_{232}	$-9-6x_0-x_0^2-2x_0x_{67}-2x_0x_{280}-6x_{67}-x_{67}^2-2x_{67}x_{280}-6x_{280}-x_{280}^2+x_{281}$
R_{233}	$-1-x_{211}x_{281}+x_{277}^2$
R_{234}	$-x_{279}^2+x_{282}$
R_{235}	$-1-x_{197}x_{282}+x_{278}^2+x_{282}$
R_{236}	$-x_{185}x_{283}-x_{277}+x_{278}$
R_{237}	$-3-x_0-4x_0x_{284}-x_{67}-4x_{67}x_{284}+x_{279}-4x_{280}x_{284}-12x_{284}$
R_{238}	$1+x_{64}+x_{65}-x_{74}+x_{263}-x_{288}$
R_{239}	$9+3x_0+x_0x_{64}+x_0x_{65}-x_0x_{178}-x_0x_{179}+x_0x_{263}+3x_{64}+x_{64}x_{67}+x_{64}x_{280}+3x_{65}+x_{65}x_{67}+x_{65}x_{280}$ $+3x_{67}-x_{67}x_{178}-x_{67}x_{179}+x_{67}x_{263}-x_{74}-x_{74}x_{273}-x_{74}x_{274}-3x_{178}-x_{178}x_{280}-3x_{179}-x_{179}x_{280}$ $+3x_{263}+x_{263}x_{280}-x_{275}x_{285}+x_{277}+3x_{280}-x_{286}$
R_{240}	$x_{273}+x_{274}-x_{286}-x_{287}$
R_{241}	$-1-2x_{29}-x_{29}^2-2x_{29}x_{292}-2x_{292}-x_{292}^2+x_{293}$
R_{242}	$-1-x_{211}x_{293}+x_{289}^2$
R_{243}	$-x_{291}^2+x_{294}$
R_{244}	$-1-x_{197}x_{294}+x_{290}^2+x_{294}$
R_{245}	$-x_{185}x_{295}-x_{289}+x_{290}$
R_{246}	$-1-x_{29}-4x_{29}x_{296}+x_{291}-4x_{292}x_{296}-4x_{296}$
R_{247}	$-x_{76}-x_{76}x_{298}-x_{76}x_{299}+x_{300}$
R_{248}	$3+3x_{29}+x_{29}x_{64}+x_{29}x_{65}-x_{29}x_{178}-x_{29}x_{179}+x_{29}x_{263}+x_{64}+x_{64}x_{292}+x_{65}+x_{65}x_{292}-x_{178}-x_{178}x_{292}$ $-x_{179}-x_{179}x_{292}+x_{263}+x_{263}x_{292}+x_{289}+3x_{292}-x_{297}x_{300}-x_{298}$
R_{249}	$-1-2x_{30}-x_{30}^2-2x_{30}x_{304}-2x_{304}-x_{304}^2+x_{305}$
R_{250}	$-1-x_{211}x_{305}+x_{301}^2$
R_{251}	$-x_{303}^2+x_{306}$
R_{252}	$-1-x_{197}x_{306}+x_{302}^2+x_{306}$
R_{253}	$-x_{185}x_{307}-x_{301}+x_{302}$
R_{254}	$-1-x_{30}-4x_{30}x_{308}+x_{303}-4x_{304}x_{308}-4x_{308}$
R_{255}	$3+3x_{30}+x_{30}x_{64}+x_{30}x_{65}-x_{30}x_{178}-x_{30}x_{179}+x_{30}x_{263}+x_{64}+x_{64}x_{304}+x_{65}+x_{65}x_{304}-x_{178}-x_{178}x_{304}$ $-x_{179}-x_{179}x_{304}+x_{263}+x_{263}x_{304}-x_{300}x_{309}+x_{301}+3x_{304}-x_{310}$
R_{256}	$x_{298}+x_{299}-x_{310}-x_{311}$
R_{257}	$-1-2x_{31}-x_{31}^2-2x_{31}x_{315}-2x_{315}-x_{315}^2+x_{316}$
R_{258}	$-1-x_{211}x_{316}+x_{312}^2$
R_{259}	$-x_{314}^2+x_{317}$
R_{260}	$-1-x_{197}x_{317}+x_{313}^2+x_{317}$
R_{261}	$-x_{185}x_{318}-x_{312}+x_{313}$

Residual	Polynomial (continued)
R_{262}	$-1-x_{31}-4x_{31}x_{319}+x_{314}-4x_{315}x_{319}-4x_{319}$
R_{263}	$3+3x_{31}+x_{31}x_{64}+x_{31}x_{65}-x_{31}x_{178}-x_{31}x_{179}+x_{31}x_{263}+x_{64}+x_{64}x_{315}+x_{65}+x_{65}x_{315}-x_{178}-x_{178}x_{315}$ $-x_{179}-x_{179}x_{315}+x_{263}+x_{263}x_{315}-x_{300}x_{320}+x_{312}+3x_{315}-x_{321}$
R_{264}	$x_{298}+x_{299}-x_{321}-x_{322}$
R_{265}	$-1-2x_{32}-x_{32}^2-2x_{32}x_{326}-2x_{326}-x_{326}^2+x_{327}$
R_{266}	$-1-x_{211}x_{327}+x_{323}^2$
R_{267}	$-x_{325}^2+x_{328}$
R_{268}	$-1-x_{197}x_{328}+x_{324}^2+x_{328}$
R_{269}	$-x_{185}x_{329}-x_{323}+x_{324}$
R_{270}	$-1-x_{32}-4x_{32}x_{330}+x_{325}-4x_{326}x_{330}-4x_{330}$
R_{271}	$3+3x_{32}+x_{32}x_{64}+x_{32}x_{65}-x_{32}x_{178}-x_{32}x_{179}+x_{32}x_{263}+x_{64}+x_{64}x_{326}+x_{65}+x_{65}x_{326}-x_{178}-x_{178}x_{326}$ $-x_{179}-x_{179}x_{326}+x_{263}+x_{263}x_{326}-x_{300}x_{331}+x_{323}+3x_{326}-x_{332}$
R_{272}	$x_{298}+x_{299}-x_{332}-x_{333}$
R_{273}	$-1-2x_{33}-x_{33}^2-2x_{33}x_{337}-2x_{337}-x_{337}^2+x_{338}$
R_{274}	$-1-x_{211}x_{338}+x_{334}^2$
R_{275}	$-x_{336}^2+x_{339}$
R_{276}	$-1-x_{197}x_{339}+x_{335}^2+x_{339}$
R_{277}	$-x_{185}x_{340}-x_{334}+x_{335}$
R_{278}	$-1-x_{33}-4x_{33}x_{341}+x_{336}-4x_{337}x_{341}-4x_{341}$
R_{279}	$-x_{76}-x_{76}x_{343}-x_{76}x_{344}+x_{345}$
R_{280}	$3+3x_{33}+x_{33}x_{64}+x_{33}x_{65}-x_{33}x_{178}-x_{33}x_{179}+x_{33}x_{263}+x_{64}+x_{64}x_{337}+x_{65}+x_{65}x_{337}-x_{178}-x_{178}x_{337}$ $-x_{179}-x_{179}x_{337}+x_{263}+x_{263}x_{337}+x_{334}+3x_{337}-x_{342}x_{345}-x_{343}$
R_{281}	$-1-2x_{34}-x_{34}^2-2x_{34}x_{349}-2x_{349}-x_{349}^2+x_{350}$
R_{282}	$-1-x_{211}x_{350}+x_{346}^2$
R_{283}	$-x_{348}^2+x_{351}$
R_{284}	$-1-x_{197}x_{351}+x_{347}^2+x_{351}$
R_{285}	$-x_{185}x_{352}-x_{346}+x_{347}$
R_{286}	$-1-x_{34}-4x_{34}x_{353}+x_{348}-4x_{349}x_{353}-4x_{353}$
R_{287}	$3+3x_{34}+x_{34}x_{64}+x_{34}x_{65}-x_{34}x_{178}-x_{34}x_{179}+x_{34}x_{263}+x_{64}+x_{64}x_{349}+x_{65}+x_{65}x_{349}-x_{178}-x_{178}x_{349}$ $-x_{179}-x_{179}x_{349}+x_{263}+x_{263}x_{349}-x_{345}x_{354}+x_{346}+3x_{349}-x_{355}$
R_{288}	$x_{343}+x_{344}-x_{355}-x_{356}$
R_{289}	$-1-2x_{35}-x_{35}^2-2x_{35}x_{360}-2x_{360}-x_{360}^2+x_{361}$
R_{290}	$-1-x_{211}x_{361}+x_{357}^2$
R_{291}	$-x_{359}^2+x_{362}$
R_{292}	$-1-x_{197}x_{362}+x_{358}^2+x_{362}$
R_{293}	$-x_{185}x_{363}-x_{357}+x_{358}$
R_{294}	$-1-x_{35}-4x_{35}x_{364}+x_{359}-4x_{360}x_{364}-4x_{364}$
R_{295}	$-x_{76}-x_{76}x_{366}-x_{76}x_{367}+x_{368}$
R_{296}	$3+3x_{35}+x_{35}x_{64}+x_{35}x_{65}-x_{35}x_{178}-x_{35}x_{179}+x_{35}x_{263}+x_{64}+x_{64}x_{360}+x_{65}+x_{65}x_{360}-x_{178}-x_{178}x_{360}$ $-x_{179}-x_{179}x_{360}+x_{263}+x_{263}x_{360}+x_{357}+3x_{360}-x_{365}x_{368}-x_{366}$
R_{297}	$-1-2x_{36}-x_{36}^2-2x_{36}x_{372}-2x_{372}-x_{372}^2+x_{373}$
R_{298}	$-1-x_{211}x_{373}+x_{369}^2$
R_{299}	$-x_{371}^2+x_{374}$
R_{300}	$-1-x_{197}x_{374}+x_{370}^2+x_{374}$
R_{301}	$-x_{185}x_{375}-x_{369}+x_{370}$
R_{302}	$-1-x_{36}-4x_{36}x_{376}+x_{371}-4x_{372}x_{376}-4x_{376}$
R_{303}	$3+3x_{36}+x_{36}x_{64}+x_{36}x_{65}-x_{36}x_{178}-x_{36}x_{179}+x_{36}x_{263}+x_{64}+x_{64}x_{372}+x_{65}+x_{65}x_{372}-x_{178}-x_{178}x_{372}$ $-x_{179}-x_{179}x_{372}+x_{263}+x_{263}x_{372}-x_{368}x_{377}+x_{369}+3x_{372}-x_{378}$
R_{304}	$x_{366}+x_{367}-x_{378}-x_{379}$
R_{305}	$-x_{195}x_{207}+x_{380}$
R_{306}	$-x_{238}x_{380}+x_{381}$
R_{307}	$-x_{247}x_{381}+x_{382}$
R_{308}	$-x_{256}x_{382}+x_{383}$

Residual	Polynomial (continued)
R_{309}	$-x_{268}x_{383}+x_{384}$
R_{310}	$-x_{281}x_{384}+x_{385}$
R_{311}	$-x_{293}x_{385}+x_{386}$
R_{312}	$-x_{305}x_{386}+x_{387}$
R_{313}	$-x_{316}x_{387}+x_{388}$
R_{314}	$-x_{327}x_{388}+x_{389}$
R_{315}	$-x_{338}x_{389}+x_{390}$
R_{316}	$-x_{350}x_{390}+x_{391}$
R_{317}	$-x_{361}x_{391}+x_{392}$
R_{318}	$-x_{373}x_{392}+x_{393}$
R_{319}	$x_{186}-x_{393}-x_{393}x_{394}$
R_{320}	$-24-3x_0-x_1-x_2-x_3-x_{26}-2x_{60}-x_{61}-2x_{62}-x_{63}-x_{64}-x_{65}-x_{66}-x_{67}-x_{68}-2x_{70}-x_{71}+x_{177}$ $-x_{261}-x_{262}-x_{263}-x_{273}-x_{274}-x_{298}-x_{299}-x_{343}-x_{344}-x_{366}-x_{367}$

B The complete 663-variable integer equation

Here all coordinates range independently over $\mathbb{Z}$. There are no additional inequalities or omitted domain predicates:

$$\boxed{Q_{663}(x_0, \ldots, x_{662}) = \sum_{j=0}^{396} S_j(x_0, \ldots, x_{662})^2 = 0.}$$

The sign conditions required by the proof have already been compiled into these residuals. The variable indices in this appendix refer only to this equation.

Residual	Polynomial (set each to zero)
S_0	$-x_4^2 + x_5$
S_1	$-x_5^2 + x_6$
S_2	$-x_6^2 + x_7$
S_3	$-x_7^2 + x_8$
S_4	$-x_8^2 + x_9$
S_5	$-x_9^2 + x_{10}$
S_6	$-x_{10}^2 + x_{11}$
S_7	$-x_{11}^2 + x_{12}$
S_8	$-x_{12}^2 + x_{13}$
S_9	$-x_{13}^2 + x_{14}$
S_{10}	$-x_{14}^2 + x_{15}$
S_{11}	$-x_{15}^2 + x_{16}$
S_{12}	$-x_{16}^2 + x_{17}$
S_{13}	$-x_{17}^2 + x_{18}$
S_{14}	$-x_{18}^2 + x_{19}$
S_{15}	$-x_{19}^2 + x_{20}$
S_{16}	$-x_{20}^2 + x_{21}$
S_{17}	$-x_{21}^2 + x_{22}$
S_{18}	$-x_{22}^2 + x_{23}$
S_{19}	$-x_{23}^2 + x_{24}$
S_{20}	$-x_{24}^2 + x_{25}$
S_{21}	$1 + x_3 - x_{25}^2$
S_{22}	$-x_0x_2 - 2x_2 + x_{28}$
S_{23}	$-2 + x_1 - x_2x_{26} - x_3 - x_{26}$
S_{24}	$-1 + x_1 - x_{27} - x_{27}x_{28}$
S_{25}	$-9 - 6x_0 - x_0^2 - 2x_0x_1 - 2x_0x_2 - 2x_0x_3 - 6x_1 - x_1^2 - 2x_1x_2 - 2x_1x_3 - 6x_2 - x_2^2 - 2x_2x_3 - 6x_3 - x_3^2 + x_{38}$
S_{26}	$-x_{38}^2 + x_{39}$
S_{27}	$-x_{39}^2 + x_{40}$
S_{28}	$-x_{40}^2 + x_{41}$
S_{29}	$-x_{41}^2 + x_{42}$
S_{30}	$-x_{42}^2 + x_{43}$
S_{31}	$-x_{43}^2 + x_{44}$
S_{32}	$-x_{44}^2 + x_{45}$
S_{33}	$-x_{45}^2 + x_{46}$
S_{34}	$-x_{46}^2 + x_{47}$
S_{35}	$-x_{47}^2 + x_{48}$
S_{36}	$-x_{48}^2 + x_{49}$
S_{37}	$-x_{49}^2 + x_{50}$
S_{38}	$-x_{50}^2 + x_{51}$
S_{39}	$-x_{51}^2 + x_{52}$

Residual	Polynomial (continued)
S_{40}	$-x_{52}^2+x_{53}$
S_{41}	$-x_{53}^2+x_{54}$
S_{42}	$-x_{54}^2+x_{55}$
S_{43}	$-x_{55}^2+x_{56}$
S_{44}	$-x_{56}^2+x_{57}$
S_{45}	$-x_{57}^2+x_{58}$
S_{46}	$-x_{58}^2+x_{59}$
S_{47}	$-x_{59}^2+x_{207}-x_{210}$
S_{48}	$-x_0x_{60}-2x_{60}+x_{64}$
S_{49}	$-x_{66}x_{67}+x_{71}$
S_{50}	$-x_{70}x_{71}+x_{72}x_{74}+x_{73}$
S_{51}	$1-2x_{60}-x_{60}x_{65}+x_{72}$
S_{52}	$-1-x_{37}x_{60}-x_{60}+x_{73}$
S_{53}	$-6-5x_{29}-x_{29}^2-2x_{29}x_{30}-5x_{30}-x_{30}^2+x_{75}$
S_{54}	$-72-68x_{30}-16x_{30}^2-16x_{30}x_{75}-34x_{75}-4x_{75}^2+x_{76}$
S_{55}	$-30-11x_{29}-x_{29}^2-4x_{29}x_{76}-22x_{76}-4x_{76}^2+x_{77}$
S_{56}	$-380-312x_{76}-64x_{76}^2-32x_{76}x_{77}-78x_{77}-4x_{77}^2+x_{78}$
S_{57}	$-6-5x_{30}-x_{30}^2-2x_{30}x_{31}-5x_{31}-x_{31}^2+x_{79}$
S_{58}	$-30-11x_{29}-x_{29}^2-8x_{29}x_{31}-4x_{29}x_{79}-44x_{31}-16x_{31}^2-16x_{31}x_{79}-22x_{79}-4x_{79}^2+x_{80}$
S_{59}	$-6-5x_{29}-x_{29}^2-2x_{29}x_{31}-5x_{31}-x_{31}^2+x_{81}$
S_{60}	$-156-100x_{30}-16x_{30}^2-32x_{30}x_{31}-16x_{30}x_{75}-16x_{30}x_{81}-100x_{31}-16x_{31}^2-16x_{31}x_{75}-16x_{31}x_{81}$ $-50x_{75}-4x_{75}^2-8x_{75}x_{81}-50x_{81}-4x_{81}^2+x_{82}$
S_{61}	$-1806-2720x_{31}-1024x_{31}^2-512x_{31}x_{79}-128x_{31}x_{80}-512x_{31}x_{81}-128x_{31}x_{82}-680x_{79}-64x_{79}^2$ $-32x_{79}x_{80}-128x_{79}x_{81}-32x_{79}x_{82}-170x_{80}-4x_{80}^2-32x_{80}x_{81}-8x_{80}x_{82}-680x_{81}-64x_{81}^2-32x_{81}x_{82}$ $-170x_{82}-4x_{82}^2+x_{83}$
S_{62}	$-110-84x_{29}-16x_{29}^2-32x_{29}x_{30}-16x_{29}x_{79}-16x_{29}x_{81}-84x_{30}-16x_{30}^2-16x_{30}x_{79}-16x_{30}x_{81}-42x_{79}$ $-4x_{79}^2-8x_{79}x_{81}-42x_{81}-4x_{81}^2+x_{84}$
S_{63}	$-756-1100x_{30}-400x_{30}^2-80x_{30}x_{75}-320x_{30}x_{79}-80x_{30}x_{84}-110x_{75}-4x_{75}^2-32x_{75}x_{79}-8x_{75}x_{84}$ $-440x_{79}-64x_{79}^2-32x_{79}x_{84}-110x_{84}-4x_{84}^2+x_{85}$
S_{64}	$-56-15x_{29}-x_{29}^2-8x_{29}x_{31}-4x_{29}x_{79}-60x_{31}-16x_{31}^2-16x_{31}x_{79}-30x_{79}-4x_{79}^2+x_{86}$
S_{65}	$-56-15x_{30}-x_{30}^2-8x_{30}x_{31}-4x_{30}x_{81}-60x_{31}-16x_{31}^2-16x_{31}x_{81}-30x_{81}-4x_{81}^2+x_{87}$
S_{66}	$-2756-3360x_{31}-1024x_{31}^2-512x_{31}x_{79}-512x_{31}x_{81}-128x_{31}x_{86}-128x_{31}x_{87}-840x_{79}-64x_{79}^2$ $-128x_{79}x_{81}-32x_{79}x_{86}-32x_{79}x_{87}-840x_{81}-64x_{81}^2-32x_{81}x_{86}-32x_{81}x_{87}-210x_{86}-4x_{86}^2-8x_{86}x_{87}$ $-210x_{87}-4x_{87}^2+x_{88}$
S_{67}	$-110-168x_{30}-64x_{30}^2-32x_{30}x_{75}-32x_{30}x_{79}-42x_{75}-4x_{75}^2-8x_{75}x_{79}-42x_{79}-4x_{79}^2+x_{89}$
S_{68}	$-756-220x_{29}-16x_{29}^2-128x_{29}x_{30}-64x_{29}x_{75}-16x_{29}x_{81}-16x_{29}x_{89}-880x_{30}-256x_{30}^2-256x_{30}x_{75}$ $-64x_{30}x_{81}-64x_{30}x_{89}-440x_{75}-64x_{75}^2-32x_{75}x_{81}-32x_{75}x_{89}-110x_{81}-4x_{81}^2-8x_{81}x_{89}-110x_{89}$ $-4x_{89}^2+x_{90}$
S_{69}	$-56-60x_{30}-16x_{30}^2-16x_{30}x_{75}-30x_{75}-4x_{75}^2+x_{91}$
S_{70}	$-30-11x_{29}-x_{29}^2-4x_{29}x_{91}-22x_{91}-4x_{91}^2+x_{92}$
S_{71}	$-380-312x_{91}-64x_{91}^2-32x_{91}x_{92}-78x_{92}-4x_{92}^2+x_{93}$
S_{72}	$-30-11x_{30}-x_{30}^2-4x_{30}x_{93}-22x_{93}-4x_{93}^2+x_{94}$
S_{73}	$-156-200x_{30}-64x_{30}^2-64x_{30}x_{75}-100x_{75}-16x_{75}^2+x_{95}$
S_{74}	$-1406-1500x_{30}-400x_{30}^2-320x_{30}x_{75}-80x_{30}x_{79}-80x_{30}x_{95}-600x_{75}-64x_{75}^2-32x_{75}x_{79}-32x_{75}x_{95}$ $-150x_{79}-4x_{79}^2-8x_{79}x_{95}-150x_{95}-4x_{95}^2+x_{96}$
S_{75}	$-16256-1020x_{29}-16x_{29}^2-512x_{29}x_{30}-256x_{29}x_{75}-16x_{29}x_{81}-64x_{29}x_{95}-16x_{29}x_{96}-16320x_{30}$ $-4096x_{30}^2-4096x_{30}x_{75}-256x_{30}x_{81}-1024x_{30}x_{95}-256x_{30}x_{96}-8160x_{75}-1024x_{75}^2-128x_{75}x_{81}$ $-512x_{75}x_{95}-128x_{75}x_{96}-510x_{81}-4x_{81}^2-32x_{81}x_{95}-8x_{81}x_{96}-2040x_{95}-64x_{95}^2-32x_{95}x_{96}-510x_{96}$ $-4x_{96}^2+x_{97}$
S_{76}	$-6-5x_{31}-x_{31}^2-2x_{31}x_{32}-5x_{32}-x_{32}^2+x_{98}$
S_{77}	$-30-11x_{29}-x_{29}^2-8x_{29}x_{32}-4x_{29}x_{98}-44x_{32}-16x_{32}^2-16x_{32}x_{98}-22x_{98}-4x_{98}^2+x_{99}$
S_{78}	$-72-136x_{31}-64x_{31}^2-32x_{31}x_{79}-32x_{31}x_{81}-34x_{79}-4x_{79}^2-8x_{79}x_{81}-34x_{81}-4x_{81}^2+x_{100}$

Residual	Polynomial (continued)
S_{79}	$-420-164x_{29}-16x_{29}^2-128x_{29}x_{31}-64x_{29}x_{79}-16x_{29}x_{99}-16x_{29}x_{100}-656x_{31}-256x_{31}^2-256x_{31}x_{79}$ $-64x_{31}x_{99}-64x_{31}x_{100}-328x_{79}-64x_{79}^2-32x_{79}x_{99}-32x_{79}x_{100}-82x_{99}-4x_{99}^2-8x_{99}x_{100}-82x_{100}$ $-4x_{100}^2+x_{101}$
S_{80}	$-462-688x_{30}-256x_{30}^2-32x_{30}x_{31}-256x_{30}x_{79}-64x_{30}x_{84}-43x_{31}-x_{31}^2-16x_{31}x_{79}-4x_{31}x_{84}$ $-344x_{79}-64x_{79}^2-32x_{79}x_{84}-86x_{84}-4x_{84}^2+x_{102}$
S_{81}	$-42-52x_{31}-16x_{31}^2-16x_{31}x_{79}-26x_{79}-4x_{79}^2+x_{103}$
S_{82}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{103}-22x_{103}-4x_{103}^2+x_{104}$
S_{83}	$-306-35x_{30}-x_{30}^2-16x_{30}x_{103}-4x_{30}x_{104}-280x_{103}-64x_{103}^2-32x_{103}x_{104}-70x_{104}-4x_{104}^2+x_{105}$
S_{84}	$-4556-4320x_{103}-1024x_{103}^2-512x_{103}x_{104}-128x_{103}x_{105}-1080x_{104}-64x_{104}^2-32x_{104}x_{105}-270x_{105}$ $-4x_{105}^2+x_{106}$
S_{85}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{106}-22x_{106}-4x_{106}^2+x_{107}$
S_{86}	$-306-35x_{30}-x_{30}^2-16x_{30}x_{106}-4x_{30}x_{107}-280x_{106}-64x_{106}^2-32x_{106}x_{107}-70x_{107}-4x_{107}^2+x_{108}$
S_{87}	$-22052-19008x_{30}-4096x_{30}^2-4096x_{30}x_{79}-1024x_{30}x_{84}-256x_{30}x_{102}-4096x_{30}x_{106}-1024x_{30}x_{107}$ $-256x_{30}x_{108}-9504x_{79}-1024x_{79}^2-512x_{79}x_{84}-128x_{79}x_{102}-2048x_{79}x_{106}-512x_{79}x_{107}-128x_{79}x_{108}$ $-2376x_{84}-64x_{84}^2-32x_{84}x_{102}-512x_{84}x_{106}-128x_{84}x_{107}-32x_{84}x_{108}-594x_{102}-4x_{102}^2-128x_{102}x_{106}$ $-32x_{102}x_{107}-8x_{102}x_{108}-9504x_{106}-1024x_{106}^2-512x_{106}x_{107}-128x_{106}x_{108}-2376x_{107}-64x_{107}^2$ $-32x_{107}x_{108}-594x_{108}-4x_{108}^2+x_{109}$
S_{88}	$-70490-531x_{29}-x_{29}^2-256x_{29}x_{106}-64x_{29}x_{107}-16x_{29}x_{108}-4x_{29}x_{109}-67968x_{106}-16384x_{106}^2$ $-8192x_{106}x_{107}-2048x_{106}x_{108}-512x_{106}x_{109}-16992x_{107}-1024x_{107}^2-512x_{107}x_{108}-128x_{107}x_{109}$ $-4248x_{108}-64x_{108}^2-32x_{108}x_{109}-1062x_{109}-4x_{109}^2+x_{110}$
S_{89}	$-1122540-1084928x_{106}-262144x_{106}^2-131072x_{106}x_{107}-32768x_{106}x_{108}-8192x_{106}x_{109}$ $-2048x_{106}x_{110}-271232x_{107}-16384x_{107}^2-8192x_{107}x_{108}-2048x_{107}x_{109}-512x_{107}x_{110}-67808x_{108}$ $-1024x_{108}^2-512x_{108}x_{109}-128x_{108}x_{110}-16952x_{109}-64x_{109}^2-32x_{109}x_{110}-4238x_{110}-4x_{110}^2+x_{111}$
S_{90}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{111}-22x_{111}-4x_{111}^2+x_{112}$
S_{91}	$-306-35x_{29}-x_{29}^2-16x_{29}x_{111}-4x_{29}x_{112}-280x_{111}-64x_{111}^2-32x_{111}x_{112}-70x_{112}-4x_{112}^2+x_{113}$
S_{92}	$-4556-4320x_{111}-1024x_{111}^2-512x_{111}x_{112}-128x_{111}x_{113}-1080x_{112}-64x_{112}^2-32x_{112}x_{113}-270x_{113}$ $-4x_{113}^2+x_{114}$
S_{93}	$-90-76x_{31}-16x_{31}^2-16x_{31}x_{79}-16x_{31}x_{114}-38x_{79}-4x_{79}^2-8x_{79}x_{114}-38x_{114}-4x_{114}^2+x_{115}$
S_{94}	$-306-35x_{30}-x_{30}^2-16x_{30}x_{114}-4x_{30}x_{115}-280x_{114}-64x_{114}^2-32x_{114}x_{115}-70x_{115}-4x_{115}^2+x_{116}$
S_{95}	$-4556-4320x_{106}-1024x_{106}^2-512x_{106}x_{107}-128x_{106}x_{108}-1080x_{107}-64x_{107}^2-32x_{107}x_{108}-270x_{108}$ $-4x_{108}^2+x_{117}$
S_{96}	$-4970-4512x_{114}-1024x_{114}^2-512x_{114}x_{115}-128x_{114}x_{116}-128x_{114}x_{117}-1128x_{115}-64x_{115}^2$ $-32x_{115}x_{116}-32x_{115}x_{117}-282x_{116}-4x_{116}^2-8x_{116}x_{117}-282x_{117}-4x_{117}^2+x_{118}$
S_{97}	$-306-35x_{31}-x_{31}^2-16x_{31}x_{117}-4x_{31}x_{118}-280x_{117}-64x_{117}^2-32x_{117}x_{118}-70x_{118}-4x_{118}^2+x_{119}$
S_{98}	$-42-52x_{30}-16x_{30}^2-8x_{30}x_{31}-16x_{30}x_{75}-13x_{31}-x_{31}^2-4x_{31}x_{75}-26x_{75}-4x_{75}^2+x_{120}$
S_{99}	$-756-1100x_{30}-400x_{30}^2-400x_{30}x_{75}-80x_{30}x_{120}-550x_{75}-100x_{75}^2-40x_{75}x_{120}-110x_{120}-4x_{120}^2$ $+x_{121}$
S_{100}	$-23870-19776x_{30}-4096x_{30}^2-4096x_{30}x_{75}-4096x_{30}x_{117}-1024x_{30}x_{118}-256x_{30}x_{119}-1024x_{30}x_{120}$ $-256x_{30}x_{121}-9888x_{75}-1024x_{75}^2-2048x_{75}x_{117}-512x_{75}x_{118}-128x_{75}x_{119}-512x_{75}x_{120}-128x_{75}x_{121}$ $-9888x_{117}-1024x_{117}^2-512x_{117}x_{118}-128x_{117}x_{119}-512x_{117}x_{120}-128x_{117}x_{121}-2472x_{118}-64x_{118}^2$ $-32x_{118}x_{119}-128x_{118}x_{120}-32x_{118}x_{121}-618x_{119}-4x_{119}^2-32x_{119}x_{120}-8x_{119}x_{121}-2472x_{120}$ $-64x_{120}^2-32x_{120}x_{121}-618x_{121}-4x_{121}^2+x_{122}$
S_{101}	$-70490-67968x_{117}-16384x_{117}^2-8192x_{117}x_{118}-2048x_{117}x_{119}-512x_{117}x_{122}-16992x_{118}-1024x_{118}^2$ $-512x_{118}x_{119}-128x_{118}x_{122}-4248x_{119}-64x_{119}^2-32x_{119}x_{122}-1062x_{122}-4x_{122}^2+x_{123}$
S_{102}	$-1-x_3-x_3x_{34}-x_{34}+x_{124}$
S_{103}	$-x_{33}^2+2x_{33}x_{34}-x_{34}^2+x_{125}$
S_{104}	$-2x_{78}x_{124}-4x_{124}+x_{126}$
S_{105}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{126}-x_{34}^2-2x_{34}x_{126}-x_{126}^2+x_{127}$
S_{106}	$-64x_{31}x_{124}-32x_{81}x_{124}-8x_{82}x_{124}-2x_{83}x_{124}-96x_{124}+x_{128}$
S_{107}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{128}-x_{34}^2-2x_{34}x_{128}-x_{128}^2+x_{129}$
S_{108}	$-64x_{30}x_{124}-32x_{79}x_{124}-8x_{84}x_{124}-2x_{85}x_{124}-80x_{124}+x_{130}$
S_{109}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{130}-x_{34}^2-2x_{34}x_{130}-x_{130}^2+x_{131}$
S_{110}	$-64x_{31}x_{124}-32x_{81}x_{124}-8x_{87}x_{124}-2x_{88}x_{124}-104x_{124}+x_{132}$

Residual	Polynomial (continued)
S_{111}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{132}-x_{34}^2-2x_{34}x_{132}-x_{132}^2+x_{133}$
S_{112}	$-64x_{30}x_{124}-32x_{75}x_{124}-8x_{89}x_{124}-2x_{90}x_{124}-80x_{124}+x_{134}$
S_{113}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{134}-x_{34}^2-2x_{34}x_{134}-x_{134}^2+x_{135}$
S_{114}	$-4x_{30}x_{124}-2x_{94}x_{124}-4x_{124}+x_{136}$
S_{115}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{136}-x_{34}^2-2x_{34}x_{136}-x_{136}^2+x_{137}$
S_{116}	$-256x_{30}x_{124}-128x_{75}x_{124}-32x_{95}x_{124}-8x_{96}x_{124}-2x_{97}x_{124}-480x_{124}+x_{138}$
S_{117}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{138}-x_{34}^2-2x_{34}x_{138}-x_{138}^2+x_{139}$
S_{118}	$-64x_{31}x_{124}-32x_{79}x_{124}-8x_{100}x_{124}-2x_{101}x_{124}-64x_{124}+x_{140}$
S_{119}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{140}-x_{34}^2-2x_{34}x_{140}-x_{140}^2+x_{141}$
S_{120}	$-x_{29}^2+2x_{29}x_{31}-x_{31}^2+x_{142}$
S_{121}	$-x_{30}^2+2x_{30}x_{31}-x_{31}^2+x_{143}$
S_{122}	$-x_{142}x_{143}+x_{144}$
S_{123}	$-4-4x_{32}-x_{32}^2+x_{145}$
S_{124}	$-x_{144}^2+2x_{144}x_{145}-x_{145}^2+x_{146}$
S_{125}	$-2x_{123}x_{124}-4x_{124}+x_{147}$
S_{126}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{147}-x_{34}^2-2x_{34}x_{147}-x_{147}^2+x_{148}$
S_{127}	$-1-x_3-x_3x_{30}-x_{30}+x_{149}$
S_{128}	$-1-x_{34}-x_{34}x_{149}-x_{149}+x_{150}$
S_{129}	$-1-x_3-x_3x_{29}-x_{29}+x_{151}$
S_{130}	$-4-4x_3-4x_3x_{30}-2x_3x_{75}-4x_{30}-2x_{75}+x_{152}$
S_{131}	$-1-x_{33}-x_{33}x_{151}-x_{151}+x_{153}$
S_{132}	$-x_{152}x_{153}-x_{153}+x_{154}$
S_{133}	$-x_{150}^2+2x_{150}x_{154}-x_{154}^2+x_{155}$
S_{134}	$-6-6x_3-4x_3x_{30}-2x_3x_{75}-4x_{30}-2x_{75}+x_{156}$
S_{135}	$-1-x_{34}-x_{34}x_{156}-x_{156}+x_{157}$
S_{136}	$-1-x_{33}-x_{33}x_{149}-x_{149}+x_{158}$
S_{137}	$-x_{157}^2+2x_{157}x_{158}-x_{158}^2+x_{159}$
S_{138}	$-x_{125}x_{127}+x_{160}$
S_{139}	$-x_{129}x_{160}+x_{161}$
S_{140}	$-x_{131}x_{161}+x_{162}$
S_{141}	$-x_{133}x_{162}+x_{163}$
S_{142}	$-x_{135}x_{163}+x_{164}$
S_{143}	$-x_{137}x_{164}+x_{165}$
S_{144}	$-x_{139}x_{165}+x_{166}$
S_{145}	$-x_{141}x_{166}+x_{167}$
S_{146}	$-x_{146}x_{167}-x_{148}x_{167}+x_{168}$
S_{147}	$-x_{155}x_{168}+x_{169}$
S_{148}	$-x_2-x_2x_{37}+x_{170}$
S_{149}	$-1-x_{35}-x_{35}x_{170}-x_{170}+x_{171}$
S_{150}	$-2+4x_1-2x_1^2+2x_1x_{33}+2x_1x_{34}+2x_1x_{171}+2x_1x_{172}-2x_{33}-x_{33}^2-2x_{33}x_{171}-2x_{34}-x_{34}^2-2x_{34}x_{172}+x_{73}x_{173}-x_{159}x_{169}-2x_{171}-x_{171}^2-2x_{172}-x_{172}^2$
S_{151}	$-1-x_2-x_2x_{36}-x_{36}-x_{36}x_{170}-x_{170}+x_{172}$
S_{152}	$1+x_{62}-x_{63}x_{73}-x_{73}+x_{265}$
S_{153}	$-x_{62}x_{174}-x_{174}-x_{174}x_{265}+x_{179}$
S_{154}	$-x_{180}-x_{180}x_{183}+x_{184}$
S_{155}	$-x_{184}^2+x_{185}$
S_{156}	$1-x_{181}^2-x_{185}+x_{185}x_{186}$
S_{157}	$-1-2x_{180}-x_{180}^2+x_{186}$
S_{158}	$5+4x_{62}+x_{62}^2-2x_{62}x_{181}-2x_{62}x_{182}+2x_{62}x_{265}+x_{179}x_{187}-4x_{181}-2x_{181}x_{265}-4x_{182}-2x_{182}x_{265}+4x_{265}+x_{265}^2$
S_{159}	$-x_{189}^2+x_{192}$
S_{160}	$1-x_{188}^2+x_{192}x_{215}$

Residual	Polynomial (continued)
S_{161}	$1 - x_{181}^2 - 2x_{181}x_{182} - x_{182}^2 + x_{215}$
S_{162}	$- 4x_{189} - 4x_{189}x_{190} + x_{193}$
S_{163}	$1 - x_{181} - x_{182} - x_{188}x_{191} + x_{193}$
S_{164}	$- x_{195}^2 + x_{198}$
S_{165}	$1 - x_{194}^2 + x_{198}x_{215}$
S_{166}	$- x_{197}^2 + x_{199}$
S_{167}	$1 - x_{196}^2 - x_{199} + x_{199}x_{200}$
S_{168}	$- 1 - 2x_{193} - x_{193}^2 + x_{200}$
S_{169}	$- x_{188}x_{201} - x_{194} + x_{196}$
S_{170}	$- 1 - x_0 - 4x_{195}x_{202} + x_{197}$
S_{171}	$5 + 4x_0 + x_0^2 - 2x_0x_{181} - 2x_0x_{182} - 4x_{181} - 4x_{182} + x_{203}$
S_{172}	$- x_0x_{195} + x_4 - x_{194} - 2x_{195} + x_{203}x_{204} + x_{205}$
S_{173}	$- x_{181}x_{195} - x_{182}x_{195} + x_{205}$
S_{174}	$- x_{207}^2 + x_{211}$
S_{175}	$1 - x_{206}^2 + x_{211}x_{215}$
S_{176}	$- x_{209}^2 + x_{212}$
S_{177}	$1 + x_{200}x_{212} - x_{208}^2 - x_{212}$
S_{178}	$- x_{188}x_{213} - x_{206} + x_{208}$
S_{179}	$- x_{207} - 4x_{207}x_{214} + x_{209} + x_{210}$
S_{180}	$x_{61} + x_{181}x_{217} + x_{181}x_{218} + x_{182}x_{217} + x_{182}x_{218} - x_{206} - 2x_{206}x_{215} - x_{207}x_{217} - x_{207}x_{218} + x_{210}x_{217} + x_{210}x_{218} - x_{215}x_{216} - 2x_{217} - 2x_{218} + x_{219}x_{220}$
S_{181}	$- 2x_{181}x_{207} - 2x_{182}x_{207} + x_{216}$
S_{182}	$- 2x_{181}x_{206} - 2x_{182}x_{206} + x_{217}$
S_{183}	$- x_{207} - 2x_{207}x_{215} + x_{218}$
S_{184}	$5 - 4x_{181} - 2x_{181}x_{207} + 2x_{181}x_{210} - 4x_{182} - 2x_{182}x_{207} + 2x_{182}x_{210} + 4x_{207} + x_{207}^2 - 2x_{207}x_{210} - 4x_{210} + x_{210}^2 + x_{219}$
S_{185}	$2 + 2x_{61} + x_{61}^2 - 2x_{61}x_{181} - 2x_{61}x_{182} - 2x_{181} - 2x_{182} + x_{221}$
S_{186}	$- x_{61}x_{207} + x_{63} + x_{181}x_{207} + x_{182}x_{207} - x_{206} - x_{207} + x_{221}x_{222}$
S_{187}	$2 + 2x_{64} + x_{64}^2 - 2x_{64}x_{181} - 2x_{64}x_{182} - 2x_{181} - 2x_{182} + x_{223}$
S_{188}	$- x_{64}x_{195} + x_{65} - x_{194} - x_{195} + x_{205} + x_{223}x_{224}$
S_{189}	$1 + x_{60}^2 - 2x_{60}x_{181} - 2x_{60}x_{182} + x_{225}$
S_{190}	$- x_{60}x_{195} + x_{66} - x_{194} + x_{205} + x_{225}x_{226}$
S_{191}	$- x_0x_{228} - x_0x_{229} + x_{68} + x_{181}x_{228} + x_{181}x_{229} + x_{182}x_{228} + x_{182}x_{229} - x_{194} - 2x_{194}x_{215} - x_{215}x_{227} - 3x_{228} - 3x_{229} + x_{230}x_{231}$
S_{192}	$- 2x_{181}x_{195} - 2x_{182}x_{195} + x_{227}$
S_{193}	$- 2x_{181}x_{194} - 2x_{182}x_{194} + x_{228}$
S_{194}	$- x_{195} - 2x_{195}x_{215} + x_{229}$
S_{195}	$10 + 6x_0 + x_0^2 - 2x_0x_{181} - 2x_0x_{182} - 6x_{181} - 6x_{182} + x_{230}$
S_{196}	$2 + 2x_{68} + x_{68}^2 - 2x_{68}x_{181} - 2x_{68}x_{182} - 2x_{181} - 2x_{182} + x_{232}$
S_{197}	$- x_{68}x_{195} - x_{194} - x_{195} + x_{205} + x_{232}x_{233} + x_{441}^2 + x_{442}^2 + x_{443} + x_{443}^2$
S_{198}	$2 + 2x_{62} + x_{62}^2 - 2x_{62}x_{181} - 2x_{62}x_{182} + 2x_{62}x_{265} - 2x_{181} - 2x_{181}x_{265} - 2x_{182} - 2x_{182}x_{265} + x_{234} + 2x_{265} + x_{265}^2$
S_{199}	$- x_{62}x_{207} + x_{174} + x_{181}x_{207} + x_{182}x_{207} - x_{206} - x_{207} - x_{207}x_{265} + x_{234}x_{235}$
S_{200}	$- x_{62}x_{195} + x_{175} - x_{194} - x_{195} - x_{195}x_{265} + x_{205} + x_{234}x_{236}$
S_{201}	$- x_{62}x_{194} - x_{62}x_{205} + x_{176} + x_{181}x_{205} + x_{182}x_{205} - x_{194} - x_{194}x_{265} - x_{195}x_{215} - x_{205} - x_{205}x_{265} + x_{234}x_{237}$
S_{202}	$- x_{239}^2 + x_{242}$
S_{203}	$1 + x_{215}x_{242} - x_{238}^2$
S_{204}	$- x_{241}^2 + x_{243}$
S_{205}	$1 + x_{200}x_{243} - x_{240}^2 - x_{243}$
S_{206}	$- x_{188}x_{244} - x_{238} + x_{240}$
S_{207}	$- 1 - x_2 - 4x_{239}x_{245} + x_{241}$
S_{208}	$- x_{62}x_{239} + x_{177} + x_{181}x_{239} + x_{182}x_{239} + x_{234}x_{246} - x_{238} - x_{239} - x_{239}x_{265}$

Residual	Polynomial (continued)
S_{209}	$-x_{248}^2+x_{251}$
S_{210}	$1+x_{215}x_{251}-x_{247}^2$
S_{211}	$-x_{250}^2+x_{252}$
S_{212}	$1+x_{200}x_{252}-x_{249}^2-x_{252}$
S_{213}	$-x_{188}x_{253}-x_{247}+x_{249}$
S_{214}	$-1-x_1-4x_{248}x_{254}+x_{250}$
S_{215}	$-x_{62}x_{248}+x_{178}+x_{181}x_{248}+x_{182}x_{248}+x_{234}x_{255}-x_{247}-x_{248}-x_{248}x_{265}$
S_{216}	$-x_{257}^2+x_{260}$
S_{217}	$1+x_{215}x_{260}-x_{256}^2$
S_{218}	$-x_{259}^2+x_{261}$
S_{219}	$1+x_{200}x_{261}-x_{258}^2-x_{261}$
S_{220}	$-x_{188}x_{262}-x_{256}+x_{258}$
S_{221}	$-x_{61}-x_{207}+x_{210}-4x_{257}x_{263}+x_{259}$
S_{222}	$-x_{62}x_{174}+x_{62}x_{257}-x_{179}x_{264}-x_{181}x_{257}-x_{182}x_{257}+x_{256}+2x_{257}+x_{257}x_{265}-x_{525}^2-x_{526}^2-x_{527}-x_{527}^2$
S_{223}	$-x_{267}^2+x_{270}$
S_{224}	$1+x_{215}x_{270}-x_{266}^2$
S_{225}	$-x_{269}^2+x_{271}$
S_{226}	$1+x_{200}x_{271}-x_{268}^2-x_{271}$
S_{227}	$-x_{188}x_{272}-x_{266}+x_{268}$
S_{228}	$-1-x_0-x_{68}-4x_{267}x_{273}+x_{269}$
S_{229}	$-x_{62}x_{175}-x_{175}-x_{175}x_{265}+x_{275}$
S_{230}	$x_{62}x_{267}-x_{69}x_{175}-x_{181}x_{267}-x_{182}x_{267}+x_{265}x_{267}+x_{266}+2x_{267}-x_{274}x_{275}-x_{540}^2-x_{541}^2-x_{542}-x_{542}^2$
S_{231}	$-x_{277}^2+x_{280}$
S_{232}	$1+x_{215}x_{280}-x_{276}^2$
S_{233}	$-x_{279}^2+x_{281}$
S_{234}	$1+x_{200}x_{281}-x_{278}^2-x_{281}$
S_{235}	$-x_{188}x_{282}-x_{276}+x_{278}$
S_{236}	$-3-x_0-x_{65}-4x_{277}x_{283}+x_{279}$
S_{237}	$x_{62}x_{277}-x_{70}x_{175}-x_{181}x_{277}-x_{182}x_{277}+x_{265}x_{277}-x_{275}x_{284}+x_{276}+2x_{277}-x_{555}^2-x_{556}^2-x_{557}-x_{557}^2$
S_{238}	$-x_{286}^2+x_{289}$
S_{239}	$1+x_{215}x_{289}-x_{285}^2$
S_{240}	$-x_{288}^2+x_{290}$
S_{241}	$1+x_{200}x_{290}-x_{287}^2-x_{290}$
S_{242}	$-x_{188}x_{291}-x_{285}+x_{287}$
S_{243}	$-1-x_{29}-4x_{286}x_{292}+x_{288}$
S_{244}	$-x_{73}x_{176}+x_{294}$
S_{245}	$-x_{62}x_{286}+x_{181}x_{286}+x_{182}x_{286}-x_{265}x_{286}-x_{285}-2x_{286}+x_{293}x_{294}+x_{570}^2+x_{571}^2+x_{572}+x_{572}^2$
S_{246}	$-x_{296}^2+x_{299}$
S_{247}	$1+x_{215}x_{299}-x_{295}^2$
S_{248}	$-x_{298}^2+x_{300}$
S_{249}	$1+x_{200}x_{300}-x_{297}^2-x_{300}$
S_{250}	$-x_{188}x_{301}-x_{295}+x_{297}$
S_{251}	$-1-x_{30}-4x_{296}x_{302}+x_{298}$
S_{252}	$-x_{62}x_{296}+x_{181}x_{296}+x_{182}x_{296}-x_{265}x_{296}+x_{294}x_{303}-x_{295}-2x_{296}+x_{582}^2+x_{583}^2+x_{584}+x_{584}^2$
S_{253}	$-x_{305}^2+x_{308}$
S_{254}	$1+x_{215}x_{308}-x_{304}^2$
S_{255}	$-x_{307}^2+x_{309}$
S_{256}	$1+x_{200}x_{309}-x_{306}^2-x_{309}$
S_{257}	$-x_{188}x_{310}-x_{304}+x_{306}$
S_{258}	$-1-x_{31}-4x_{305}x_{311}+x_{307}$

Residual	Polynomial (continued)
S_{259}	$-x_{62}x_{305}+x_{181}x_{305}+x_{182}x_{305}-x_{265}x_{305}+x_{294}x_{312}-x_{304}-2x_{305}+x_{594}^2+x_{595}^2+x_{596}+x_{596}^2$
S_{260}	$-x_{314}^2+x_{317}$
S_{261}	$1+x_{215}x_{317}-x_{313}^2$
S_{262}	$-x_{316}^2+x_{318}$
S_{263}	$1+x_{200}x_{318}-x_{315}^2-x_{318}$
S_{264}	$-x_{188}x_{319}-x_{313}+x_{315}$
S_{265}	$-1-x_{32}-4x_{314}x_{320}+x_{316}$
S_{266}	$-x_{62}x_{314}+x_{181}x_{314}+x_{182}x_{314}-x_{265}x_{314}+x_{294}x_{321}-x_{313}-2x_{314}+x_{606}^2+x_{607}^2+x_{608}+x_{608}^2$
S_{267}	$-x_{323}^2+x_{326}$
S_{268}	$1+x_{215}x_{326}-x_{322}^2$
S_{269}	$-x_{325}^2+x_{327}$
S_{270}	$1+x_{200}x_{327}-x_{324}^2-x_{327}$
S_{271}	$-x_{188}x_{328}-x_{322}+x_{324}$
S_{272}	$-1-x_{33}-4x_{323}x_{329}+x_{325}$
S_{273}	$-x_{73}x_{177}+x_{331}$
S_{274}	$-x_{62}x_{323}+x_{181}x_{323}+x_{182}x_{323}-x_{265}x_{323}-x_{322}-2x_{323}+x_{330}x_{331}+x_{618}^2+x_{619}^2+x_{620}+x_{620}^2$
S_{275}	$-x_{333}^2+x_{336}$
S_{276}	$1+x_{215}x_{336}-x_{332}^2$
S_{277}	$-x_{335}^2+x_{337}$
S_{278}	$1+x_{200}x_{337}-x_{334}^2-x_{337}$
S_{279}	$-x_{188}x_{338}-x_{332}+x_{334}$
S_{280}	$-1-x_{34}-4x_{333}x_{339}+x_{335}$
S_{281}	$-x_{62}x_{333}+x_{181}x_{333}+x_{182}x_{333}-x_{265}x_{333}+x_{331}x_{340}-x_{332}-2x_{333}+x_{630}^2+x_{631}^2+x_{632}+x_{632}^2$
S_{282}	$-x_{342}^2+x_{345}$
S_{283}	$1+x_{215}x_{345}-x_{341}^2$
S_{284}	$-x_{344}^2+x_{346}$
S_{285}	$1+x_{200}x_{346}-x_{343}^2-x_{346}$
S_{286}	$-x_{188}x_{347}-x_{341}+x_{343}$
S_{287}	$-1-x_{35}-4x_{342}x_{348}+x_{344}$
S_{288}	$-x_{73}x_{178}+x_{350}$
S_{289}	$-x_{62}x_{342}+x_{181}x_{342}+x_{182}x_{342}-x_{265}x_{342}-x_{341}-2x_{342}+x_{349}x_{350}+x_{642}^2+x_{643}^2+x_{644}+x_{644}^2$
S_{290}	$-x_{352}^2+x_{355}$
S_{291}	$1+x_{215}x_{355}-x_{351}^2$
S_{292}	$-x_{354}^2+x_{356}$
S_{293}	$1+x_{200}x_{356}-x_{353}^2-x_{356}$
S_{294}	$-x_{188}x_{357}-x_{351}+x_{353}$
S_{295}	$-1-x_{36}-4x_{352}x_{358}+x_{354}$
S_{296}	$-x_{62}x_{352}+x_{181}x_{352}+x_{182}x_{352}-x_{265}x_{352}+x_{350}x_{359}-x_{351}-2x_{352}+x_{654}^2+x_{655}^2+x_{656}+x_{656}^2$
S_{297}	$-x_{198}x_{211}+x_{360}$
S_{298}	$-x_{242}x_{360}+x_{361}$
S_{299}	$-x_{251}x_{361}+x_{362}$
S_{300}	$-x_{260}x_{362}+x_{363}$
S_{301}	$-x_{270}x_{363}+x_{364}$
S_{302}	$-x_{280}x_{364}+x_{365}$
S_{303}	$-x_{289}x_{365}+x_{366}$
S_{304}	$-x_{299}x_{366}+x_{367}$
S_{305}	$-x_{308}x_{367}+x_{368}$
S_{306}	$-x_{317}x_{368}+x_{369}$
S_{307}	$-x_{326}x_{369}+x_{370}$
S_{308}	$-x_{336}x_{370}+x_{371}$
S_{309}	$-x_{345}x_{371}+x_{372}$

Residual	Polynomial (continued)
S_{310}	$-x_{355}x_{372}+x_{373}$
S_{311}	$x_{189}-x_{373}-x_{373}x_{374}$
S_{312}	$-16-3x_0-x_1-x_2-x_4-x_{60}-2x_{61}-x_{62}-x_{63}-x_{64}-x_{65}-x_{66}-2x_{68}-x_{174}-x_{175}-x_{176}-x_{177}-x_{178}+x_{180}-2x_{207}+2x_{210}-x_{265}-x_{441}^2-x_{442}^2-x_{443}-x_{443}^2$
S_{313}	$x_0-x_{375}^2-x_{376}^2-x_{377}-x_{377}^2$
S_{314}	$x_3-x_{378}^2-x_{379}^2-x_{380}-x_{380}^2$
S_{315}	$x_4-x_{381}^2-x_{382}^2-x_{383}-x_{383}^2$
S_{316}	$-2+x_2-x_3-x_{384}^2-x_{385}^2-x_{386}-x_{386}^2$
S_{317}	$x_{27}-x_{387}^2-x_{388}^2-x_{389}-x_{389}^2$
S_{318}	$x_{29}-x_{390}^2-x_{391}^2-x_{392}-x_{392}^2$
S_{319}	$x_{30}-x_{393}^2-x_{394}^2-x_{395}-x_{395}^2$
S_{320}	$x_{31}-x_{396}^2-x_{397}^2-x_{398}-x_{398}^2$
S_{321}	$x_{32}-x_{399}^2-x_{400}^2-x_{401}-x_{401}^2$
S_{322}	$x_{33}-x_{402}^2-x_{403}^2-x_{404}-x_{404}^2$
S_{323}	$x_{34}-x_{405}^2-x_{406}^2-x_{407}-x_{407}^2$
S_{324}	$x_{35}-x_{408}^2-x_{409}^2-x_{410}-x_{410}^2$
S_{325}	$x_{36}-x_{411}^2-x_{412}^2-x_{413}-x_{413}^2$
S_{326}	$x_{37}-x_{414}^2-x_{415}^2-x_{416}-x_{416}^2$
S_{327}	$x_{61}-x_{417}^2-x_{418}^2-x_{419}-x_{419}^2$
S_{328}	$x_{63}-x_{420}^2-x_{421}^2-x_{422}-x_{422}^2$
S_{329}	$-1-x_{60}x_{62}+x_{62}+x_{63}-x_{423}^2-x_{424}^2-x_{425}-x_{425}^2$
S_{330}	$x_{60}x_{62}-x_{63}-x_{426}^2-x_{427}^2-x_{428}-x_{428}^2$
S_{331}	$x_{65}-x_{429}^2-x_{430}^2-x_{431}-x_{431}^2$
S_{332}	$x_{66}-x_{432}^2-x_{433}^2-x_{434}-x_{434}^2$
S_{333}	$x_{67}-x_{435}^2-x_{436}^2-x_{437}-x_{437}^2$
S_{334}	$x_{68}-x_{438}^2-x_{439}^2-x_{440}-x_{440}^2$
S_{335}	$-1-x_{67}x_{69}+x_{69}+x_{441}^2+x_{442}^2+x_{443}+x_{443}^2-x_{444}^2-x_{445}^2-x_{446}-x_{446}^2$
S_{336}	$x_{67}x_{69}-x_{441}^2-x_{442}^2-x_{443}-x_{443}^2-x_{447}^2-x_{448}^2-x_{449}-x_{449}^2$
S_{337}	$x_{70}-x_{450}^2-x_{451}^2-x_{452}-x_{452}^2$
S_{338}	$-1+x_{72}-x_{73}-x_{453}^2-x_{454}^2-x_{455}-x_{455}^2$
S_{339}	$-8-x_{29}-x_{30}-x_{31}-x_{32}-x_{33}-x_{34}-x_{35}-x_{36}+8x_{73}-x_{456}^2-x_{457}^2-x_{458}-x_{458}^2$
S_{340}	$x_{181}-x_{459}^2-x_{460}^2-x_{461}-x_{461}^2$
S_{341}	$x_{182}-x_{462}^2-x_{463}^2-x_{464}-x_{464}^2$
S_{342}	$x_{183}-x_{465}^2-x_{466}^2-x_{467}-x_{467}^2$
S_{343}	$x_{190}-x_{468}^2-x_{469}^2-x_{470}-x_{470}^2$
S_{344}	$x_{194}-x_{471}^2-x_{472}^2-x_{473}-x_{473}^2$
S_{345}	$-1-x_0+x_{195}-x_{474}^2-x_{475}^2-x_{476}-x_{476}^2$
S_{346}	$x_{203}-x_{477}^2-x_{478}^2-x_{479}-x_{479}^2$
S_{347}	$x_{206}-x_{480}^2-x_{481}^2-x_{482}-x_{482}^2$
S_{348}	$x_{210}-x_{483}^2-x_{484}^2-x_{485}-x_{485}^2$
S_{349}	$x_{219}-x_{486}^2-x_{487}^2-x_{488}-x_{488}^2$
S_{350}	$x_{221}-x_{489}^2-x_{490}^2-x_{491}-x_{491}^2$
S_{351}	$x_{223}-x_{492}^2-x_{493}^2-x_{494}-x_{494}^2$
S_{352}	$x_{225}-x_{495}^2-x_{496}^2-x_{497}-x_{497}^2$
S_{353}	$x_{230}-x_{498}^2-x_{499}^2-x_{500}-x_{500}^2$
S_{354}	$x_{232}-x_{501}^2-x_{502}^2-x_{503}-x_{503}^2$
S_{355}	$x_{234}-x_{504}^2-x_{505}^2-x_{506}-x_{506}^2$
S_{356}	$x_{238}-x_{507}^2-x_{508}^2-x_{509}-x_{509}^2$
S_{357}	$-1-x_2+x_{239}-x_{510}^2-x_{511}^2-x_{512}-x_{512}^2$
S_{358}	$x_{247}-x_{513}^2-x_{514}^2-x_{515}-x_{515}^2$

Residual	Polynomial (continued)
S_{359}	$-1-x_{1}+x_{248}-x_{516}^2-x_{517}^2-x_{518}-x_{518}^2$
S_{360}	$x_{256}-x_{519}^2-x_{520}^2-x_{521}-x_{521}^2$
S_{361}	$-x_{61}-x_{207}+x_{210}+x_{257}-x_{522}^2-x_{523}^2-x_{524}-x_{524}^2$
S_{362}	$-1+x_{174}-x_{525}^2-x_{526}^2-x_{527}-x_{527}^2-x_{528}^2-x_{529}^2-x_{530}-x_{530}^2$
S_{363}	$x_{265}-x_{531}^2-x_{532}^2-x_{533}-x_{533}^2$
S_{364}	$x_{266}-x_{534}^2-x_{535}^2-x_{536}-x_{536}^2$
S_{365}	$-1-x_{0}-x_{68}+x_{267}-x_{537}^2-x_{538}^2-x_{539}-x_{539}^2$
S_{366}	$-1+x_{175}-x_{540}^2-x_{541}^2-x_{542}-x_{542}^2-x_{543}^2-x_{544}^2-x_{545}-x_{545}^2$
S_{367}	$x_{62}-x_{69}+x_{265}-x_{546}^2-x_{547}^2-x_{548}-x_{548}^2$
S_{368}	$x_{276}-x_{549}^2-x_{550}^2-x_{551}-x_{551}^2$
S_{369}	$-3-x_{0}-x_{65}+x_{277}-x_{552}^2-x_{553}^2-x_{554}-x_{554}^2$
S_{370}	$-1+x_{175}-x_{555}^2-x_{556}^2-x_{557}-x_{557}^2-x_{558}^2-x_{559}^2-x_{560}-x_{560}^2$
S_{371}	$x_{62}-x_{70}+x_{265}-x_{561}^2-x_{562}^2-x_{563}-x_{563}^2$
S_{372}	$x_{285}-x_{564}^2-x_{565}^2-x_{566}-x_{566}^2$
S_{373}	$-1-x_{29}+x_{286}-x_{567}^2-x_{568}^2-x_{569}-x_{569}^2$
S_{374}	$-1+x_{176}-x_{570}^2-x_{571}^2-x_{572}-x_{572}^2-x_{573}^2-x_{574}^2-x_{575}-x_{575}^2$
S_{375}	$x_{295}-x_{576}^2-x_{577}^2-x_{578}-x_{578}^2$
S_{376}	$-1-x_{30}+x_{296}-x_{579}^2-x_{580}^2-x_{581}-x_{581}^2$
S_{377}	$-1+x_{176}-x_{582}^2-x_{583}^2-x_{584}-x_{584}^2-x_{585}^2-x_{586}^2-x_{587}-x_{587}^2$
S_{378}	$x_{304}-x_{588}^2-x_{589}^2-x_{590}-x_{590}^2$
S_{379}	$-1-x_{31}+x_{305}-x_{591}^2-x_{592}^2-x_{593}-x_{593}^2$
S_{380}	$-1+x_{176}-x_{594}^2-x_{595}^2-x_{596}-x_{596}^2-x_{597}^2-x_{598}^2-x_{599}-x_{599}^2$
S_{381}	$x_{313}-x_{600}^2-x_{601}^2-x_{602}-x_{602}^2$
S_{382}	$-1-x_{32}+x_{314}-x_{603}^2-x_{604}^2-x_{605}-x_{605}^2$
S_{383}	$-1+x_{176}-x_{606}^2-x_{607}^2-x_{608}-x_{608}^2-x_{609}^2-x_{610}^2-x_{611}-x_{611}^2$
S_{384}	$x_{322}-x_{612}^2-x_{613}^2-x_{614}-x_{614}^2$
S_{385}	$-1-x_{33}+x_{323}-x_{615}^2-x_{616}^2-x_{617}-x_{617}^2$
S_{386}	$-1+x_{177}-x_{618}^2-x_{619}^2-x_{620}-x_{620}^2-x_{621}^2-x_{622}^2-x_{623}-x_{623}^2$
S_{387}	$x_{332}-x_{624}^2-x_{625}^2-x_{626}-x_{626}^2$
S_{388}	$-1-x_{34}+x_{333}-x_{627}^2-x_{628}^2-x_{629}-x_{629}^2$
S_{389}	$-1+x_{177}-x_{630}^2-x_{631}^2-x_{632}-x_{632}^2-x_{633}^2-x_{634}^2-x_{635}-x_{635}^2$
S_{390}	$x_{341}-x_{636}^2-x_{637}^2-x_{638}-x_{638}^2$
S_{391}	$-1-x_{35}+x_{342}-x_{639}^2-x_{640}^2-x_{641}-x_{641}^2$
S_{392}	$-1+x_{178}-x_{642}^2-x_{643}^2-x_{644}-x_{644}^2-x_{645}^2-x_{646}^2-x_{647}-x_{647}^2$
S_{393}	$x_{351}-x_{648}^2-x_{649}^2-x_{650}-x_{650}^2$
S_{394}	$-1-x_{36}+x_{352}-x_{651}^2-x_{652}^2-x_{653}-x_{653}^2$
S_{395}	$-1+x_{178}-x_{654}^2-x_{655}^2-x_{656}-x_{656}^2-x_{657}^2-x_{658}^2-x_{659}-x_{659}^2$
S_{396}	$x_{374}-x_{660}^2-x_{661}^2-x_{662}-x_{662}^2$

C The complete 687-variable small-coefficient equation

Again each coordinate ranges independently over $\mathbb{Z}$. The equation is

$$\boxed{Q_{687}(x_0,\dots,x_{686}) = \sum_{j=0}^{420} T_j(x_0,\dots,x_{686})^2 = 0.}$$

Its fully collected coefficient height is 63,906,013. It trades a few additional coordinates for fewer expanded terms and much smaller coefficients than the principal integer variant.

Residual	Polynomial (set each to zero)
T_0	$-x_4^2 + x_5$
T_1	$-x_5^2 + x_6$
T_2	$-x_6^2 + x_7$
T_3	$-x_7^2 + x_8$
T_4	$-x_8^2 + x_9$
T_5	$-x_9^2 + x_{10}$
T_6	$-x_{10}^2 + x_{11}$
T_7	$-x_{11}^2 + x_{12}$
T_8	$-x_{12}^2 + x_{13}$
T_9	$-x_{13}^2 + x_{14}$
T_{10}	$-x_{14}^2 + x_{15}$
T_{11}	$-x_{15}^2 + x_{16}$
T_{12}	$-x_{16}^2 + x_{17}$
T_{13}	$-x_{17}^2 + x_{18}$
T_{14}	$-x_{18}^2 + x_{19}$
T_{15}	$-x_{19}^2 + x_{20}$
T_{16}	$-x_{20}^2 + x_{21}$
T_{17}	$-x_{21}^2 + x_{22}$
T_{18}	$-x_{22}^2 + x_{23}$
T_{19}	$-x_{23}^2 + x_{24}$
T_{20}	$-x_{24}^2 + x_{25}$
T_{21}	$1 + x_3 - x_{25}^2$
T_{22}	$-x_0x_2 - 2x_2 + x_{28}$
T_{23}	$-2 + x_1 - x_2x_{26} - x_3 - x_{26}$
T_{24}	$-1 + x_1 - x_{27} - x_{27}x_{28}$
T_{25}	$-3 - x_0 - x_1 - x_2 - x_3 + x_{38}$
T_{26}	$-x_{38}^2 + x_{39}$
T_{27}	$-x_{39}^2 + x_{40}$
T_{28}	$-x_{40}^2 + x_{41}$
T_{29}	$-x_{41}^2 + x_{42}$
T_{30}	$-x_{42}^2 + x_{43}$
T_{31}	$-x_{43}^2 + x_{44}$
T_{32}	$-x_{44}^2 + x_{45}$
T_{33}	$-x_{45}^2 + x_{46}$
T_{34}	$-x_{46}^2 + x_{47}$
T_{35}	$-x_{47}^2 + x_{48}$
T_{36}	$-x_{48}^2 + x_{49}$
T_{37}	$-x_{49}^2 + x_{50}$
T_{38}	$-x_{50}^2 + x_{51}$
T_{39}	$-x_{51}^2 + x_{52}$

Residual	Polynomial (continued)
T_{40}	$- x_{52}^2 + x_{53}$
T_{41}	$- x_{53}^2 + x_{54}$
T_{42}	$- x_{54}^2 + x_{55}$
T_{43}	$- x_{55}^2 + x_{56}$
T_{44}	$- x_{56}^2 + x_{57}$
T_{45}	$- x_{57}^2 + x_{58}$
T_{46}	$- x_{58}^2 + x_{59}$
T_{47}	$- x_{59}^2 + x_{60}$
T_{48}	$- x_{60}^2 + x_{222} - x_{225}$
T_{49}	$- x_0 x_{61} - 2x_{61} + x_{65}$
T_{50}	$- x_{67} x_{68} + x_{72}$
T_{51}	$- x_{71} x_{72} + x_{73} x_{75} + x_{74}$
T_{52}	$1 - 2x_{61} - x_{61} x_{66} + x_{73}$
T_{53}	$- 1 - x_{37} x_{61} - x_{61} + x_{74}$
T_{54}	$- 6 - 5x_{29} - x_{29}^2 - 2x_{29} x_{30} - 5x_{30} - x_{30}^2 + x_{76}$
T_{55}	$- 72 - 68x_{30} - 16x_{30}^2 - 16x_{30} x_{76} - 34x_{76} - 4x_{76}^2 + x_{77}$
T_{56}	$- 30 - 11x_{29} - x_{29}^2 - 4x_{29} x_{77} - 22x_{77} - 4x_{77}^2 + x_{78}$
T_{57}	$- 380 - 312x_{77} - 64x_{77}^2 - 32x_{77} x_{78} - 78x_{78} - 4x_{78}^2 + x_{79}$
T_{58}	$- 6 - 5x_{30} - x_{30}^2 - 2x_{30} x_{31} - 5x_{31} - x_{31}^2 + x_{80}$
T_{59}	$- 30 - 11x_{29} - x_{29}^2 - 8x_{29} x_{31} - 4x_{29} x_{80} - 44x_{31} - 16x_{31}^2 - 16x_{31} x_{80} - 22x_{80} - 4x_{80}^2 + x_{81}$
T_{60}	$- 6 - 5x_{29} - x_{29}^2 - 2x_{29} x_{31} - 5x_{31} - x_{31}^2 + x_{82}$
T_{61}	$- 12 - 4x_{30} - 4x_{31} - 2x_{76} - 2x_{82} + x_{83}$
T_{62}	$- 42 - 32x_{31} - 8x_{80} - 2x_{81} - 8x_{82} - 2x_{83} - 2x_{83}^2 + x_{84}$
T_{63}	$- 96 - 64x_{31} - 32x_{82} - 8x_{83} - 8x_{83}^2 - 2x_{84} - 2x_{84}^2 + x_{85}$
T_{64}	$- 10 - 4x_{29} - 4x_{30} - 2x_{80} - 2x_{82} + x_{86}$
T_{65}	$- 27 - 20x_{30} - 2x_{76} - 8x_{80} - 2x_{86} - 2x_{86}^2 + x_{87}$
T_{66}	$- 80 - 64x_{30} - 32x_{80} - 8x_{86} - 8x_{86}^2 - 2x_{87} - 2x_{87}^2 + x_{88}$
T_{67}	$- 56 - 15x_{29} - x_{29}^2 - 8x_{29} x_{31} - 4x_{29} x_{80} - 60x_{31} - 16x_{31}^2 - 16x_{31} x_{80} - 30x_{80} - 4x_{80}^2 + x_{89}$
T_{68}	$- 56 - 15x_{30} - x_{30}^2 - 8x_{30} x_{31} - 4x_{30} x_{82} - 60x_{31} - 16x_{31}^2 - 16x_{31} x_{82} - 30x_{82} - 4x_{82}^2 + x_{90}$
T_{69}	$- 52 - 32x_{31} - 8x_{80} - 8x_{82} - 2x_{89} - 2x_{90} + x_{91}$
T_{70}	$- 104 - 64x_{31} - 32x_{82} - 8x_{90} - 2x_{91} - 2x_{91}^2 + x_{92}$
T_{71}	$- 110 - 168x_{30} - 64x_{30}^2 - 32x_{30} x_{76} - 32x_{30} x_{80} - 42x_{76} - 4x_{76}^2 - 8x_{76} x_{80} - 42x_{80} - 4x_{80}^2 + x_{93}$
T_{72}	$- 27 - 4x_{29} - 16x_{30} - 8x_{76} - 2x_{82} - 2x_{93} + x_{94}$
T_{73}	$- 80 - 64x_{30} - 32x_{76} - 8x_{93} - 2x_{94} - 2x_{94}^2 + x_{95}$
T_{74}	$- 56 - 60x_{30} - 16x_{30}^2 - 16x_{30} x_{76} - 30x_{76} - 4x_{76}^2 + x_{96}$
T_{75}	$- 30 - 11x_{29} - x_{29}^2 - 4x_{29} x_{96} - 22x_{96} - 4x_{96}^2 + x_{97}$
T_{76}	$- 380 - 312x_{96} - 64x_{96}^2 - 32x_{96} x_{97} - 78x_{97} - 4x_{97}^2 + x_{98}$
T_{77}	$- 30 - 11x_{30} - x_{30}^2 - 4x_{30} x_{98} - 22x_{98} - 4x_{98}^2 + x_{99}$
T_{78}	$- 156 - 200x_{30} - 64x_{30}^2 - 64x_{30} x_{76} - 100x_{76} - 16x_{76}^2 + x_{100}$
T_{79}	$- 37 - 20x_{30} - 8x_{76} - 2x_{80} - 2x_{100} + x_{101}$
T_{80}	$- 120 - 64x_{30} - 32x_{76} - 8x_{100} - 2x_{101} - 2x_{101}^2 + x_{102}$
T_{81}	$- 56 - 60x_{29} - 16x_{29}^2 - 16x_{29} x_{82} - 8x_{29} x_{102} - 30x_{82} - 4x_{82}^2 - 4x_{82} x_{102} - 15x_{102} - x_{102}^2 + x_{103}$
T_{82}	$- 6 - 5x_{31} - x_{31}^2 - 2x_{31} x_{32} - 5x_{32} - x_{32}^2 + x_{104}$
T_{83}	$- 30 - 11x_{29} - x_{29}^2 - 8x_{29} x_{32} - 4x_{29} x_{104} - 44x_{32} - 16x_{32}^2 - 16x_{32} x_{104} - 22x_{104} - 4x_{104}^2 + x_{105}$
T_{84}	$- 72 - 136x_{31} - 64x_{31}^2 - 32x_{31} x_{80} - 32x_{31} x_{82} - 34x_{80} - 4x_{80}^2 - 8x_{80} x_{82} - 34x_{82} - 4x_{82}^2 + x_{106}$
T_{85}	$- 20 - 4x_{29} - 16x_{31} - 8x_{80} - 2x_{105} - 2x_{106} + x_{107}$
T_{86}	$- 64 - 64x_{31} - 32x_{80} - 8x_{106} - 2x_{107} - 2x_{107}^2 + x_{108}$
T_{87}	$- 21 - 16x_{30} - x_{31} - 8x_{80} - 2x_{86} - 2x_{86}^2 + x_{109}$
T_{88}	$- 80 - 64x_{30} - 32x_{80} - 8x_{86} - 8x_{86}^2 - 2x_{109} - 2x_{109}^2 + x_{110}$
T_{89}	$- 42 - 52x_{31} - 16x_{31}^2 - 16x_{31} x_{80} - 26x_{80} - 4x_{80}^2 + x_{111}$

Residual	Polynomial (continued)
T_{90}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{111}-22x_{111}-4x_{111}^2+x_{112}$
T_{91}	$-306-35x_{30}-x_{30}^2-16x_{30}x_{111}-4x_{30}x_{112}-280x_{111}-64x_{111}^2-32x_{111}x_{112}-70x_{112}-4x_{112}^2+x_{113}$
T_{92}	$-4556-4320x_{111}-1024x_{111}^2-512x_{111}x_{112}-128x_{111}x_{113}-1080x_{112}-64x_{112}^2-32x_{112}x_{113}-270x_{113}-4x_{113}^2+x_{114}$
T_{93}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{114}-22x_{114}-4x_{114}^2+x_{115}$
T_{94}	$-306-35x_{30}-x_{30}^2-16x_{30}x_{114}-4x_{30}x_{115}-280x_{114}-64x_{114}^2-32x_{114}x_{115}-70x_{115}-4x_{115}^2+x_{116}$
T_{95}	$-68-x_{110}-32x_{114}-8x_{115}-2x_{116}+x_{117}$
T_{96}	$-264-128x_{114}-32x_{115}-8x_{116}-2x_{117}-2x_{117}^2+x_{118}$
T_{97}	$-2-3x_{29}-x_{29}^2-2x_{29}x_{118}-3x_{118}-x_{118}^2+x_{119}$
T_{98}	$-12-28x_{118}-16x_{118}^2-16x_{118}x_{119}-14x_{119}-4x_{119}^2+x_{120}$
T_{99}	$-30-11x_{31}-x_{31}^2-4x_{31}x_{120}-22x_{120}-4x_{120}^2+x_{121}$
T_{100}	$-306-35x_{29}-x_{29}^2-16x_{29}x_{120}-4x_{29}x_{121}-280x_{120}-64x_{120}^2-32x_{120}x_{121}-70x_{121}-4x_{121}^2+x_{122}$
T_{101}	$-4556-4320x_{120}-1024x_{120}^2-512x_{120}x_{121}-128x_{120}x_{122}-1080x_{121}-64x_{121}^2-32x_{121}x_{122}-270x_{122}-4x_{122}^2+x_{123}$
T_{102}	$-90-76x_{31}-16x_{31}^2-16x_{31}x_{80}-16x_{31}x_{123}-38x_{80}-4x_{80}^2-8x_{80}x_{123}-38x_{123}-4x_{123}^2+x_{124}$
T_{103}	$-306-35x_{30}-x_{30}^2-16x_{30}x_{123}-4x_{30}x_{124}-280x_{123}-64x_{123}^2-32x_{123}x_{124}-70x_{124}-4x_{124}^2+x_{125}$
T_{104}	$-4556-4320x_{114}-1024x_{114}^2-512x_{114}x_{115}-128x_{114}x_{116}-1080x_{115}-64x_{115}^2-32x_{115}x_{116}-270x_{116}-4x_{116}^2+x_{126}$
T_{105}	$-70-32x_{123}-8x_{124}-2x_{125}-2x_{126}+x_{127}$
T_{106}	$-x_{127}-x_{127}^2+x_{128}$
T_{107}	$-306-35x_{31}-x_{31}^2-16x_{31}x_{126}-4x_{31}x_{128}-280x_{126}-64x_{126}^2-32x_{126}x_{128}-70x_{128}-4x_{128}^2+x_{129}$
T_{108}	$-42-52x_{30}-16x_{30}^2-8x_{30}x_{31}-16x_{30}x_{76}-13x_{31}-x_{31}^2-4x_{31}x_{76}-26x_{76}-4x_{76}^2+x_{130}$
T_{109}	$-756-1100x_{30}-400x_{30}^2-400x_{30}x_{76}-80x_{30}x_{130}-550x_{76}-100x_{76}^2-40x_{76}x_{130}-110x_{130}-4x_{130}^2+x_{131}$
T_{110}	$-88-64x_{30}-32x_{76}-8x_{130}-2x_{131}+x_{132}$
T_{111}	$-66-32x_{126}-8x_{128}-2x_{129}-x_{132}+x_{133}$
T_{112}	$-264-128x_{126}-32x_{128}-8x_{129}-2x_{133}-2x_{133}^2+x_{134}$
T_{113}	$-2-3x_{134}-x_{134}^2+x_{135}$
T_{114}	$-1-x_3-x_3x_{34}-x_{34}+x_{136}$
T_{115}	$-x_{33}^2+2x_{33}x_{34}-x_{34}^2+x_{137}$
T_{116}	$-2x_{79}x_{136}-4x_{136}+x_{138}$
T_{117}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{138}-x_{34}^2-2x_{34}x_{138}-x_{138}^2+x_{139}$
T_{118}	$-x_{85}x_{136}+x_{140}$
T_{119}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{140}-x_{34}^2-2x_{34}x_{140}-x_{140}^2+x_{141}$
T_{120}	$-x_{88}x_{136}+x_{142}$
T_{121}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{142}-x_{34}^2-2x_{34}x_{142}-x_{142}^2+x_{143}$
T_{122}	$-x_{92}x_{136}+x_{144}$
T_{123}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{144}-x_{34}^2-2x_{34}x_{144}-x_{144}^2+x_{145}$
T_{124}	$-x_{95}x_{136}+x_{146}$
T_{125}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{146}-x_{34}^2-2x_{34}x_{146}-x_{146}^2+x_{147}$
T_{126}	$-4x_{30}x_{136}-2x_{99}x_{136}-4x_{136}+x_{148}$
T_{127}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{148}-x_{34}^2-2x_{34}x_{148}-x_{148}^2+x_{149}$
T_{128}	$-4x_{102}x_{136}-2x_{103}x_{136}+x_{150}$
T_{129}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{150}-x_{34}^2-2x_{34}x_{150}-x_{150}^2+x_{151}$
T_{130}	$-x_{108}x_{136}+x_{152}$
T_{131}	$-x_{33}^2+2x_{33}x_{34}+2x_{33}x_{152}-x_{34}^2-2x_{34}x_{152}-x_{152}^2+x_{153}$
T_{132}	$-x_{29}^2+2x_{29}x_{31}-x_{31}^2+x_{154}$
T_{133}	$-x_{30}^2+2x_{30}x_{31}-x_{31}^2+x_{155}$
T_{134}	$-x_{154}x_{155}+x_{156}$
T_{135}	$-4-4x_{32}-x_{32}^2+x_{157}$
T_{136}	$-x_{156}^2+2x_{156}x_{157}-x_{157}^2+x_{158}$

Residual	Polynomial (continued)
T_{137}	$-2x_{135}x_{136} - 4x_{136} + x_{159}$
T_{138}	$-x_{33}^2 + 2x_{33}x_{34} + 2x_{33}x_{159} - x_{34}^2 - 2x_{34}x_{159} - x_{159}^2 + x_{160}$
T_{139}	$-1 - x_3 - x_3x_{30} - x_{30} + x_{161}$
T_{140}	$-1 - x_{34} - x_{34}x_{161} - x_{161} + x_{162}$
T_{141}	$-1 - x_3 - x_3x_{29} - x_{29} + x_{163}$
T_{142}	$-4 - 4x_3 - 4x_3x_{30} - 2x_3x_{76} - 4x_{30} - 2x_{76} + x_{164}$
T_{143}	$-1 - x_{33} - x_{33}x_{163} - x_{163} + x_{165}$
T_{144}	$-x_{164}x_{165} - x_{165} + x_{166}$
T_{145}	$-x_{162}^2 + 2x_{162}x_{166} - x_{166}^2 + x_{167}$
T_{146}	$-6 - 6x_3 - 4x_3x_{30} - 2x_3x_{76} - 4x_{30} - 2x_{76} + x_{168}$
T_{147}	$-1 - x_{34} - x_{34}x_{168} - x_{168} + x_{169}$
T_{148}	$-1 - x_{33} - x_{33}x_{161} - x_{161} + x_{170}$
T_{149}	$-x_{169}^2 + 2x_{169}x_{170} - x_{170}^2 + x_{171}$
T_{150}	$-x_{137}x_{139} + x_{172}$
T_{151}	$-x_{141}x_{172} + x_{173}$
T_{152}	$-x_{143}x_{173} + x_{174}$
T_{153}	$-x_{145}x_{174} + x_{175}$
T_{154}	$-x_{147}x_{175} + x_{176}$
T_{155}	$-x_{149}x_{176} + x_{177}$
T_{156}	$-x_{151}x_{177} + x_{178}$
T_{157}	$-x_{153}x_{178} + x_{179}$
T_{158}	$-x_{158}x_{179} - x_{160}x_{179} + x_{180}$
T_{159}	$-x_{167}x_{180} + x_{181}$
T_{160}	$-x_2 - x_2x_{37} + x_{182}$
T_{161}	$-1 - x_{35} - x_{35}x_{182} - x_{182} + x_{183}$
T_{162}	$-1 + 2x_1 - x_1^2 + 2x_1x_{33} + 2x_1x_{183} - 2x_{33} - x_{33}^2 - 2x_{33}x_{183} + x_{74}x_{186} - x_{171}x_{181} - 2x_{183} - x_{183}^2 - x_{185}$
T_{163}	$-1 - x_2 - x_2x_{36} - x_{36} - x_{36}x_{182} - x_{182} + x_{184}$
T_{164}	$-1 + 2x_1 - x_1^2 + 2x_1x_{34} + 2x_1x_{184} - 2x_{34} - x_{34}^2 - 2x_{34}x_{184} - 2x_{184} - x_{184}^2 + x_{185}$
T_{165}	$-4 - x_{29} - x_{30} - x_{31} - x_{32} + x_{187}$
T_{166}	$-4 - x_{33} - x_{34} - x_{35} - x_{36} + 8x_{74} - x_{187} - x_{480}^2 - x_{481}^2 - x_{482} - x_{482}^2$
T_{167}	$1 + x_{63} - x_{64}x_{74} - x_{74} + x_{284}$
T_{168}	$-x_{63}x_{188} - x_{188} - x_{188}x_{284} + x_{193}$
T_{169}	$-x_{194} - x_{194}x_{197} + x_{198}$
T_{170}	$-x_{198}^2 + x_{199}$
T_{171}	$1 - x_{195}^2 - x_{199} + x_{199}x_{200}$
T_{172}	$-1 - 2x_{194} - x_{194}^2 + x_{200}$
T_{173}	$1 - 2x_{63}x_{195} - 2x_{63}x_{196} + x_{193}x_{201} - 4x_{195} - 2x_{195}x_{284} - 4x_{196} - 2x_{196}x_{284} + x_{202}$
T_{174}	$-4 - 4x_{63} - x_{63}^2 - 2x_{63}x_{284} + x_{202} - 4x_{284} - x_{284}^2$
T_{175}	$-x_{204}^2 + x_{207}$
T_{176}	$1 - x_{203}^2 + x_{207}x_{230}$
T_{177}	$1 - x_{195}^2 - 2x_{195}x_{196} - x_{196}^2 + x_{230}$
T_{178}	$-4x_{204} - 4x_{204}x_{205} + x_{208}$
T_{179}	$1 - x_{195} - x_{196} - x_{203}x_{206} + x_{208}$
T_{180}	$-x_{210}^2 + x_{213}$
T_{181}	$1 - x_{209}^2 + x_{213}x_{230}$
T_{182}	$-x_{212}^2 + x_{214}$
T_{183}	$1 - x_{211}^2 - x_{214} + x_{214}x_{215}$
T_{184}	$-1 - 2x_{208} - x_{208}^2 + x_{215}$
T_{185}	$-x_{203}x_{216} - x_{209} + x_{211}$
T_{186}	$-1 - x_0 - 4x_{210}x_{217} + x_{212}$
T_{187}	$5 + 4x_0 + x_0^2 - 2x_0x_{195} - 2x_0x_{196} - 4x_{195} - 4x_{196} + x_{218}$

Residual	Polynomial (continued)
T_{188}	$-x_0x_{210}+x_4-x_{209}-2x_{210}+x_{218}x_{219}+x_{220}$
T_{189}	$-x_{195}x_{210}-x_{196}x_{210}+x_{220}$
T_{190}	$-x_{222}^2+x_{226}$
T_{191}	$1-x_{221}^2+x_{226}x_{230}$
T_{192}	$-x_{224}^2+x_{227}$
T_{193}	$1+x_{215}x_{227}-x_{223}^2-x_{227}$
T_{194}	$-x_{203}x_{228}-x_{221}+x_{223}$
T_{195}	$-x_{222}-4x_{222}x_{229}+x_{224}+x_{225}$
T_{196}	$-2x_{195}x_{222}-2x_{196}x_{222}+x_{231}$
T_{197}	$-2x_{195}x_{221}-2x_{196}x_{221}+x_{232}$
T_{198}	$-x_{222}-2x_{222}x_{230}+x_{233}$
T_{199}	$1-4x_{195}-2x_{195}x_{222}+2x_{195}x_{225}-4x_{196}-2x_{196}x_{222}+2x_{196}x_{225}+x_{234}+x_{235}$
T_{200}	$-4-4x_{222}-x_{222}^2+2x_{222}x_{225}+4x_{225}-x_{225}^2+x_{235}$
T_{201}	$-x_{222}x_{232}-x_{222}x_{233}+x_{225}x_{232}+x_{225}x_{233}-2x_{232}-2x_{233}+x_{237}$
T_{202}	$-x_{62}-x_{195}x_{232}-x_{195}x_{233}-x_{196}x_{232}-x_{196}x_{233}+x_{221}+2x_{221}x_{230}+x_{230}x_{231}-x_{234}x_{236}+x_{237}$
T_{203}	$2+2x_{62}+x_{62}^2-2x_{62}x_{195}-2x_{62}x_{196}-2x_{195}-2x_{196}+x_{238}$
T_{204}	$-x_{62}x_{222}+x_{64}+x_{195}x_{222}+x_{196}x_{222}-x_{221}-x_{222}+x_{238}x_{239}$
T_{205}	$2+2x_{65}+x_{65}^2-2x_{65}x_{195}-2x_{65}x_{196}-2x_{195}-2x_{196}+x_{240}$
T_{206}	$-x_{65}x_{210}+x_{66}-x_{209}-x_{210}+x_{220}+x_{240}x_{241}$
T_{207}	$1+x_{61}^2-2x_{61}x_{195}-2x_{61}x_{196}+x_{242}$
T_{208}	$-x_{61}x_{210}+x_{67}-x_{209}+x_{220}+x_{242}x_{243}$
T_{209}	$-2x_{195}x_{210}-2x_{196}x_{210}+x_{244}$
T_{210}	$-2x_{195}x_{209}-2x_{196}x_{209}+x_{245}$
T_{211}	$-x_{210}-2x_{210}x_{230}+x_{246}$
T_{212}	$10+6x_0+x_0^2-2x_0x_{195}-2x_0x_{196}-6x_{195}-6x_{196}+x_{247}$
T_{213}	$x_0x_{245}+x_0x_{246}-x_{69}-x_{195}x_{245}-x_{195}x_{246}-x_{196}x_{245}-x_{196}x_{246}+x_{209}+2x_{209}x_{230}+x_{230}x_{244}+3x_{245}+3x_{246}-x_{247}x_{248}$
T_{214}	$2+2x_{69}+x_{69}^2-2x_{69}x_{195}-2x_{69}x_{196}-2x_{195}-2x_{196}+x_{249}$
T_{215}	$-x_{69}x_{210}-x_{209}-x_{210}+x_{220}+x_{249}x_{250}+x_{465}^2+x_{466}^2+x_{467}+x_{467}^2$
T_{216}	$1-2x_{63}x_{195}-2x_{63}x_{196}-2x_{195}-2x_{195}x_{284}-2x_{196}-2x_{196}x_{284}+x_{251}+x_{252}$
T_{217}	$-1-2x_{63}-x_{63}^2-2x_{63}x_{284}+x_{252}-2x_{284}-x_{284}^2$
T_{218}	$-x_{63}x_{222}+x_{188}+x_{195}x_{222}+x_{196}x_{222}-x_{221}-x_{222}-x_{222}x_{284}+x_{251}x_{253}$
T_{219}	$-x_{63}x_{210}+x_{189}-x_{209}-x_{210}-x_{210}x_{284}+x_{220}+x_{251}x_{254}$
T_{220}	$-x_{63}x_{209}-x_{63}x_{220}-x_{209}-x_{209}x_{284}-x_{220}-x_{220}x_{284}+x_{256}$
T_{221}	$-x_{190}-x_{195}x_{220}-x_{196}x_{220}+x_{210}x_{230}-x_{251}x_{255}+x_{256}$
T_{222}	$-x_{258}^2+x_{261}$
T_{223}	$1+x_{230}x_{261}-x_{257}^2$
T_{224}	$-x_{260}^2+x_{262}$
T_{225}	$1+x_{215}x_{262}-x_{259}^2-x_{262}$
T_{226}	$-x_{203}x_{263}-x_{257}+x_{259}$
T_{227}	$-1-x_2-4x_{258}x_{264}+x_{260}$
T_{228}	$-x_{63}x_{258}+x_{191}+x_{195}x_{258}+x_{196}x_{258}+x_{251}x_{265}-x_{257}-x_{258}-x_{258}x_{284}$
T_{229}	$-x_{267}^2+x_{270}$
T_{230}	$1+x_{230}x_{270}-x_{266}^2$
T_{231}	$-x_{269}^2+x_{271}$
T_{232}	$1+x_{215}x_{271}-x_{268}^2-x_{271}$
T_{233}	$-x_{203}x_{272}-x_{266}+x_{268}$
T_{234}	$-1-x_1-4x_{267}x_{273}+x_{269}$
T_{235}	$-x_{63}x_{267}+x_{192}+x_{195}x_{267}+x_{196}x_{267}+x_{251}x_{274}-x_{266}-x_{267}-x_{267}x_{284}$
T_{236}	$-x_{276}^2+x_{279}$
T_{237}	$1+x_{230}x_{279}-x_{275}^2$

Residual	Polynomial (continued)
T_{238}	$-x_{278}^2+x_{280}$
T_{239}	$1+x_{215}x_{280}-x_{277}^2-x_{280}$
T_{240}	$-x_{203}x_{281}-x_{275}+x_{277}$
T_{241}	$-x_{62}-x_{222}+x_{225}-4x_{276}x_{282}+x_{278}$
T_{242}	$-x_{63}x_{188}+x_{63}x_{276}-x_{193}x_{283}-x_{195}x_{276}-x_{196}x_{276}+x_{275}+2x_{276}+x_{276}x_{284}-x_{549}^2-x_{550}^2-x_{551}-x_{551}^2$
T_{243}	$-x_{286}^2+x_{289}$
T_{244}	$1+x_{230}x_{289}-x_{285}^2$
T_{245}	$-x_{288}^2+x_{290}$
T_{246}	$1+x_{215}x_{290}-x_{287}^2-x_{290}$
T_{247}	$-x_{203}x_{291}-x_{285}+x_{287}$
T_{248}	$-1-x_0-x_{69}-4x_{286}x_{292}+x_{288}$
T_{249}	$-x_{63}x_{189}-x_{189}-x_{189}x_{284}+x_{294}$
T_{250}	$x_{63}x_{286}-x_{70}x_{189}-x_{195}x_{286}-x_{196}x_{286}+x_{284}x_{286}+x_{285}+2x_{286}-x_{293}x_{294}-x_{564}^2-x_{565}^2-x_{566}-x_{566}^2$
T_{251}	$-x_{296}^2+x_{299}$
T_{252}	$1+x_{230}x_{299}-x_{295}^2$
T_{253}	$-x_{298}^2+x_{300}$
T_{254}	$1+x_{215}x_{300}-x_{297}^2-x_{300}$
T_{255}	$-x_{203}x_{301}-x_{295}+x_{297}$
T_{256}	$-3-x_0-x_{66}-4x_{296}x_{302}+x_{298}$
T_{257}	$x_{63}x_{296}-x_{71}x_{189}-x_{195}x_{296}-x_{196}x_{296}+x_{284}x_{296}-x_{294}x_{303}+x_{295}+2x_{296}-x_{579}^2-x_{580}^2-x_{581}-x_{581}^2$
T_{258}	$-x_{305}^2+x_{308}$
T_{259}	$1+x_{230}x_{308}-x_{304}^2$
T_{260}	$-x_{307}^2+x_{309}$
T_{261}	$1+x_{215}x_{309}-x_{306}^2-x_{309}$
T_{262}	$-x_{203}x_{310}-x_{304}+x_{306}$
T_{263}	$-1-x_{29}-4x_{305}x_{311}+x_{307}$
T_{264}	$-x_{74}x_{190}+x_{313}$
T_{265}	$-x_{63}x_{305}+x_{195}x_{305}+x_{196}x_{305}-x_{284}x_{305}-x_{304}-2x_{305}+x_{312}x_{313}+x_{594}^2+x_{595}^2+x_{596}+x_{596}^2$
T_{266}	$-x_{315}^2+x_{318}$
T_{267}	$1+x_{230}x_{318}-x_{314}^2$
T_{268}	$-x_{317}^2+x_{319}$
T_{269}	$1+x_{215}x_{319}-x_{316}^2-x_{319}$
T_{270}	$-x_{203}x_{320}-x_{314}+x_{316}$
T_{271}	$-1-x_{30}-4x_{315}x_{321}+x_{317}$
T_{272}	$-x_{63}x_{315}+x_{195}x_{315}+x_{196}x_{315}-x_{284}x_{315}+x_{313}x_{322}-x_{314}-2x_{315}+x_{606}^2+x_{607}^2+x_{608}+x_{608}^2$
T_{273}	$-x_{324}^2+x_{327}$
T_{274}	$1+x_{230}x_{327}-x_{323}^2$
T_{275}	$-x_{326}^2+x_{328}$
T_{276}	$1+x_{215}x_{328}-x_{325}^2-x_{328}$
T_{277}	$-x_{203}x_{329}-x_{323}+x_{325}$
T_{278}	$-1-x_{31}-4x_{324}x_{330}+x_{326}$
T_{279}	$-x_{63}x_{324}+x_{195}x_{324}+x_{196}x_{324}-x_{284}x_{324}+x_{313}x_{331}-x_{323}-2x_{324}+x_{618}^2+x_{619}^2+x_{620}+x_{620}^2$
T_{280}	$-x_{333}^2+x_{336}$
T_{281}	$1+x_{230}x_{336}-x_{332}^2$
T_{282}	$-x_{335}^2+x_{337}$
T_{283}	$1+x_{215}x_{337}-x_{334}^2-x_{337}$
T_{284}	$-x_{203}x_{338}-x_{332}+x_{334}$
T_{285}	$-1-x_{32}-4x_{333}x_{339}+x_{335}$
T_{286}	$-x_{63}x_{333}+x_{195}x_{333}+x_{196}x_{333}-x_{284}x_{333}+x_{313}x_{340}-x_{332}-2x_{333}+x_{630}^2+x_{631}^2+x_{632}+x_{632}^2$
T_{287}	$-x_{342}^2+x_{345}$

Residual	Polynomial (continued)
T_{288}	$1 + x_{230}x_{345} - x_{341}^2$
T_{289}	$- x_{344}^2 + x_{346}$
T_{290}	$1 + x_{215}x_{346} - x_{343}^2 - x_{346}$
T_{291}	$- x_{203}x_{347} - x_{341} + x_{343}$
T_{292}	$- 1 - x_{33} - 4x_{342}x_{348} + x_{344}$
T_{293}	$- x_{74}x_{191} + x_{350}$
T_{294}	$- x_{63}x_{342} + x_{195}x_{342} + x_{196}x_{342} - x_{284}x_{342} - x_{341} - 2x_{342} + x_{349}x_{350} + x_{642}^2 + x_{643}^2 + x_{644} + x_{644}^2$
T_{295}	$- x_{352}^2 + x_{355}$
T_{296}	$1 + x_{230}x_{355} - x_{351}^2$
T_{297}	$- x_{354}^2 + x_{356}$
T_{298}	$1 + x_{215}x_{356} - x_{353}^2 - x_{356}$
T_{299}	$- x_{203}x_{357} - x_{351} + x_{353}$
T_{300}	$- 1 - x_{34} - 4x_{352}x_{358} + x_{354}$
T_{301}	$- x_{63}x_{352} + x_{195}x_{352} + x_{196}x_{352} - x_{284}x_{352} + x_{350}x_{359} - x_{351} - 2x_{352} + x_{654}^2 + x_{655}^2 + x_{656} + x_{656}^2$
T_{302}	$- x_{361}^2 + x_{364}$
T_{303}	$1 + x_{230}x_{364} - x_{360}^2$
T_{304}	$- x_{363}^2 + x_{365}$
T_{305}	$1 + x_{215}x_{365} - x_{362}^2 - x_{365}$
T_{306}	$- x_{203}x_{366} - x_{360} + x_{362}$
T_{307}	$- 1 - x_{35} - 4x_{361}x_{367} + x_{363}$
T_{308}	$- x_{74}x_{192} + x_{369}$
T_{309}	$- x_{63}x_{361} + x_{195}x_{361} + x_{196}x_{361} - x_{284}x_{361} - x_{360} - 2x_{361} + x_{368}x_{369} + x_{666}^2 + x_{667}^2 + x_{668} + x_{668}^2$
T_{310}	$- x_{371}^2 + x_{374}$
T_{311}	$1 + x_{230}x_{374} - x_{370}^2$
T_{312}	$- x_{373}^2 + x_{375}$
T_{313}	$1 + x_{215}x_{375} - x_{372}^2 - x_{375}$
T_{314}	$- x_{203}x_{376} - x_{370} + x_{372}$
T_{315}	$- 1 - x_{36} - 4x_{371}x_{377} + x_{373}$
T_{316}	$- x_{63}x_{371} + x_{195}x_{371} + x_{196}x_{371} - x_{284}x_{371} + x_{369}x_{378} - x_{370} - 2x_{371} + x_{678}^2 + x_{679}^2 + x_{680} + x_{680}^2$
T_{317}	$- x_{213}x_{226} + x_{379}$
T_{318}	$- x_{261}x_{379} + x_{380}$
T_{319}	$- x_{270}x_{380} + x_{381}$
T_{320}	$- x_{279}x_{381} + x_{382}$
T_{321}	$- x_{289}x_{382} + x_{383}$
T_{322}	$- x_{299}x_{383} + x_{384}$
T_{323}	$- x_{308}x_{384} + x_{385}$
T_{324}	$- x_{318}x_{385} + x_{386}$
T_{325}	$- x_{327}x_{386} + x_{387}$
T_{326}	$- x_{336}x_{387} + x_{388}$
T_{327}	$- x_{345}x_{388} + x_{389}$
T_{328}	$- x_{355}x_{389} + x_{390}$
T_{329}	$- x_{364}x_{390} + x_{391}$
T_{330}	$- x_{374}x_{391} + x_{392}$
T_{331}	$x_{204} - x_{392} - x_{392}x_{393}$
T_{332}	$- 9 - 3x_0 - x_2 - x_4 - x_{222} + x_{225} + x_{394}$
T_{333}	$- 2 - x_{61} - x_{62} - x_{222} + x_{225} - x_{394} + x_{395}$
T_{334}	$- 2 - x_{62} - x_{64} - x_{65} - x_{395} + x_{396}$
T_{335}	$- 1 - x_{66} - x_{67} - 2x_{69} - x_{396} + x_{397}$
T_{336}	$- 2 - x_1 - x_{63} - x_{284} - x_{397} + x_{398} - x_{465}^2 - x_{466}^2 - x_{467} - x_{467}^2$
T_{337}	$- x_{188} - x_{189} - x_{190} - x_{191} - x_{192} + x_{194} - x_{398}$
T_{338}	$x_0 - x_{399}^2 - x_{400}^2 - x_{401} - x_{401}^2$

Residual	Polynomial (continued)
T_{339}	$x_3 - x_{402}^2 - x_{403}^2 - x_{404} - x_{404}^2$
T_{340}	$x_4 - x_{405}^2 - x_{406}^2 - x_{407} - x_{407}^2$
T_{341}	$-2 + x_2 - x_3 - x_{408}^2 - x_{409}^2 - x_{410} - x_{410}^2$
T_{342}	$x_{27} - x_{411}^2 - x_{412}^2 - x_{413} - x_{413}^2$
T_{343}	$x_{29} - x_{414}^2 - x_{415}^2 - x_{416} - x_{416}^2$
T_{344}	$x_{30} - x_{417}^2 - x_{418}^2 - x_{419} - x_{419}^2$
T_{345}	$x_{31} - x_{420}^2 - x_{421}^2 - x_{422} - x_{422}^2$
T_{346}	$x_{32} - x_{423}^2 - x_{424}^2 - x_{425} - x_{425}^2$
T_{347}	$x_{33} - x_{426}^2 - x_{427}^2 - x_{428} - x_{428}^2$
T_{348}	$x_{34} - x_{429}^2 - x_{430}^2 - x_{431} - x_{431}^2$
T_{349}	$x_{35} - x_{432}^2 - x_{433}^2 - x_{434} - x_{434}^2$
T_{350}	$x_{36} - x_{435}^2 - x_{436}^2 - x_{437} - x_{437}^2$
T_{351}	$x_{37} - x_{438}^2 - x_{439}^2 - x_{440} - x_{440}^2$
T_{352}	$x_{62} - x_{441}^2 - x_{442}^2 - x_{443} - x_{443}^2$
T_{353}	$x_{64} - x_{444}^2 - x_{445}^2 - x_{446} - x_{446}^2$
T_{354}	$-1 - x_{61}x_{63} + x_{63} + x_{64} - x_{447}^2 - x_{448}^2 - x_{449} - x_{449}^2$
T_{355}	$x_{61}x_{63} - x_{64} - x_{450}^2 - x_{451}^2 - x_{452} - x_{452}^2$
T_{356}	$x_{66} - x_{453}^2 - x_{454}^2 - x_{455} - x_{455}^2$
T_{357}	$x_{67} - x_{456}^2 - x_{457}^2 - x_{458} - x_{458}^2$
T_{358}	$x_{68} - x_{459}^2 - x_{460}^2 - x_{461} - x_{461}^2$
T_{359}	$x_{69} - x_{462}^2 - x_{463}^2 - x_{464} - x_{464}^2$
T_{360}	$-1 - x_{68}x_{70} + x_{70} + x_{465}^2 + x_{466}^2 + x_{467} + x_{467}^2 - x_{468}^2 - x_{469}^2 - x_{470} - x_{470}^2$
T_{361}	$x_{68}x_{70} - x_{465}^2 - x_{466}^2 - x_{467} - x_{467}^2 - x_{471}^2 - x_{472}^2 - x_{473} - x_{473}^2$
T_{362}	$x_{71} - x_{474}^2 - x_{475}^2 - x_{476} - x_{476}^2$
T_{363}	$-1 + x_{73} - x_{74} - x_{477}^2 - x_{478}^2 - x_{479} - x_{479}^2$
T_{364}	$x_{195} - x_{483}^2 - x_{484}^2 - x_{485} - x_{485}^2$
T_{365}	$x_{196} - x_{486}^2 - x_{487}^2 - x_{488} - x_{488}^2$
T_{366}	$x_{197} - x_{489}^2 - x_{490}^2 - x_{491} - x_{491}^2$
T_{367}	$x_{205} - x_{492}^2 - x_{493}^2 - x_{494} - x_{494}^2$
T_{368}	$x_{209} - x_{495}^2 - x_{496}^2 - x_{497} - x_{497}^2$
T_{369}	$-1 - x_0 + x_{210} - x_{498}^2 - x_{499}^2 - x_{500} - x_{500}^2$
T_{370}	$x_{218} - x_{501}^2 - x_{502}^2 - x_{503} - x_{503}^2$
T_{371}	$x_{221} - x_{504}^2 - x_{505}^2 - x_{506} - x_{506}^2$
T_{372}	$x_{225} - x_{507}^2 - x_{508}^2 - x_{509} - x_{509}^2$
T_{373}	$x_{234} - x_{510}^2 - x_{511}^2 - x_{512} - x_{512}^2$
T_{374}	$x_{238} - x_{513}^2 - x_{514}^2 - x_{515} - x_{515}^2$
T_{375}	$x_{240} - x_{516}^2 - x_{517}^2 - x_{518} - x_{518}^2$
T_{376}	$x_{242} - x_{519}^2 - x_{520}^2 - x_{521} - x_{521}^2$
T_{377}	$x_{247} - x_{522}^2 - x_{523}^2 - x_{524} - x_{524}^2$
T_{378}	$x_{249} - x_{525}^2 - x_{526}^2 - x_{527} - x_{527}^2$
T_{379}	$x_{251} - x_{528}^2 - x_{529}^2 - x_{530} - x_{530}^2$
T_{380}	$x_{257} - x_{531}^2 - x_{532}^2 - x_{533} - x_{533}^2$
T_{381}	$-1 - x_2 + x_{258} - x_{534}^2 - x_{535}^2 - x_{536} - x_{536}^2$
T_{382}	$x_{266} - x_{537}^2 - x_{538}^2 - x_{539} - x_{539}^2$
T_{383}	$-1 - x_1 + x_{267} - x_{540}^2 - x_{541}^2 - x_{542} - x_{542}^2$
T_{384}	$x_{275} - x_{543}^2 - x_{544}^2 - x_{545} - x_{545}^2$
T_{385}	$-x_{62} - x_{222} + x_{225} + x_{276} - x_{546}^2 - x_{547}^2 - x_{548} - x_{548}^2$
T_{386}	$-1 + x_{188} - x_{549}^2 - x_{550}^2 - x_{551} - x_{551}^2 - x_{552}^2 - x_{553}^2 - x_{554} - x_{554}^2$
T_{387}	$x_{284} - x_{555}^2 - x_{556}^2 - x_{557} - x_{557}^2$
T_{388}	$x_{285} - x_{558}^2 - x_{559}^2 - x_{560} - x_{560}^2$

Residual	Polynomial (continued)
T_{389}	$-1-x_{0}-x_{69}+x_{286}-x_{561}^2-x_{562}^2-x_{563}-x_{563}^2$
T_{390}	$-1+x_{189}-x_{564}^2-x_{565}^2-x_{566}-x_{566}^2-x_{567}^2-x_{568}^2-x_{569}-x_{569}^2$
T_{391}	$x_{63}-x_{70}+x_{284}-x_{570}^2-x_{571}^2-x_{572}-x_{572}^2$
T_{392}	$x_{295}-x_{573}^2-x_{574}^2-x_{575}-x_{575}^2$
T_{393}	$-3-x_{0}-x_{66}+x_{296}-x_{576}^2-x_{577}^2-x_{578}-x_{578}^2$
T_{394}	$-1+x_{189}-x_{579}^2-x_{580}^2-x_{581}-x_{581}^2-x_{582}^2-x_{583}^2-x_{584}-x_{584}^2$
T_{395}	$x_{63}-x_{71}+x_{284}-x_{585}^2-x_{586}^2-x_{587}-x_{587}^2$
T_{396}	$x_{304}-x_{588}^2-x_{589}^2-x_{590}-x_{590}^2$
T_{397}	$-1-x_{29}+x_{305}-x_{591}^2-x_{592}^2-x_{593}-x_{593}^2$
T_{398}	$-1+x_{190}-x_{594}^2-x_{595}^2-x_{596}-x_{596}^2-x_{597}^2-x_{598}^2-x_{599}-x_{599}^2$
T_{399}	$x_{314}-x_{600}^2-x_{601}^2-x_{602}-x_{602}^2$
T_{400}	$-1-x_{30}+x_{315}-x_{603}^2-x_{604}^2-x_{605}-x_{605}^2$
T_{401}	$-1+x_{190}-x_{606}^2-x_{607}^2-x_{608}-x_{608}^2-x_{609}^2-x_{610}^2-x_{611}-x_{611}^2$
T_{402}	$x_{323}-x_{612}^2-x_{613}^2-x_{614}-x_{614}^2$
T_{403}	$-1-x_{31}+x_{324}-x_{615}^2-x_{616}^2-x_{617}-x_{617}^2$
T_{404}	$-1+x_{190}-x_{618}^2-x_{619}^2-x_{620}-x_{620}^2-x_{621}^2-x_{622}^2-x_{623}-x_{623}^2$
T_{405}	$x_{332}-x_{624}^2-x_{625}^2-x_{626}-x_{626}^2$
T_{406}	$-1-x_{32}+x_{333}-x_{627}^2-x_{628}^2-x_{629}-x_{629}^2$
T_{407}	$-1+x_{190}-x_{630}^2-x_{631}^2-x_{632}-x_{632}^2-x_{633}^2-x_{634}^2-x_{635}-x_{635}^2$
T_{408}	$x_{341}-x_{636}^2-x_{637}^2-x_{638}-x_{638}^2$
T_{409}	$-1-x_{33}+x_{342}-x_{639}^2-x_{640}^2-x_{641}-x_{641}^2$
T_{410}	$-1+x_{191}-x_{642}^2-x_{643}^2-x_{644}-x_{644}^2-x_{645}^2-x_{646}^2-x_{647}-x_{647}^2$
T_{411}	$x_{351}-x_{648}^2-x_{649}^2-x_{650}-x_{650}^2$
T_{412}	$-1-x_{34}+x_{352}-x_{651}^2-x_{652}^2-x_{653}-x_{653}^2$
T_{413}	$-1+x_{191}-x_{654}^2-x_{655}^2-x_{656}-x_{656}^2-x_{657}^2-x_{658}^2-x_{659}-x_{659}^2$
T_{414}	$x_{360}-x_{660}^2-x_{661}^2-x_{662}-x_{662}^2$
T_{415}	$-1-x_{35}+x_{361}-x_{663}^2-x_{664}^2-x_{665}-x_{665}^2$
T_{416}	$-1+x_{192}-x_{666}^2-x_{667}^2-x_{668}-x_{668}^2-x_{669}^2-x_{670}^2-x_{671}-x_{671}^2$
T_{417}	$x_{370}-x_{672}^2-x_{673}^2-x_{674}-x_{674}^2$
T_{418}	$-1-x_{36}+x_{371}-x_{675}^2-x_{676}^2-x_{677}-x_{677}^2$
T_{419}	$-1+x_{192}-x_{678}^2-x_{679}^2-x_{680}-x_{680}^2-x_{681}^2-x_{682}^2-x_{683}-x_{683}^2$
T_{420}	$x_{393}-x_{684}^2-x_{685}^2-x_{686}-x_{686}^2$